

CHAPTER 1

INTRODUCTION

Sustainable development and efficient management of water is an increasingly complex challenge in India. Increasing population, growing urbanisation and rapid industrialisation combined with the need for raising agricultural production generates competing claims for water. Ground water has an important role in meeting the water requirements of agriculture, industrial and domestic sectors in India. About 85 percent of India's rural domestic water requirements, 50 percent of its urban water requirements and more than 50 percent of its irrigation requirements are being met from ground water resources. Ground water is annually replenishable resource but its availability is non-uniform in space and time. Technically, the dynamic ground water refers to the quantity of ground water available in the zone of water level fluctuation, which is replenished annually. Hence, the sustainable development of ground water resources warrants precise quantitative assessment based on reasonably valid scientific principles. National Water Policy, 2012 has laid emphasis on periodic assessment of ground water resources on scientific basis. The trends in water availability due to various factors including climate change must also be assessed and accounted for during water resources planning. To meet the increasing demands of water, it advocates direct use of rainfall, desalination and avoidance of inadvertent evapo-transpiration for augmenting utilizable water resources. The National Water Policy, 2012 also states that safe water for drinking and sanitation should be considered as pre-emptive needs followed by high priority allocation for other domestic needs (including needs of animals), achieving food security, supporting sustenance agriculture and minimum eco-system needs. Available water, after meeting the above needs should be allocated in a manner to promote its conservation and efficient use.

1.1 Previous Assessments:

The assessment of water resources of the country dates back to 1901 when the First Irrigation Commission assessed the Surface Water Resources as 144 million hectare meters (M.ham) (NABARD, 2006). In 1949 Dr. A. N. Khosla based on empirical formula estimated the total average annual runoff of all the river systems of India including both surface and ground water resources as 167 M.ham (Central Ground Water Board, 1995). Since then attempts have been made from time to time by various Working Groups/ Committees/Task Forces constituted by Govt. of India to estimate the ground water resources of the country based on available data and in response to developmental needs. In 1976, the National Commission of Agriculture assessed the total ground water resources of the country as 67 M.ham. and the Utilizable ground water was worked out to be 35 M.ham, out of which 26 M.ham was considered available for irrigation (Central Ground Water Board, 1995).

The first systematic methodology to estimate the ground water resources of the country was evolved by Ground Water Over-Exploitation Committee in 1979. The committee was constituted by Agriculture Refinance and Development Corporation (ARDC) and was headed by Chairman, CGWB with Members from – State Ground Water Organizations and Financial Institutions. Based on the norms suggested by the committee, the country's Gross Ground Water Recharge has been assessed as 47 M.ham. and the Net Recharge as 32 M.ham (Central Ground Water Board, 1995).

In 1982, Government of India constituted 'Ground Water Estimation Committee' (GEC) drawing Members from various States / Central organizations engaged in hydrogeological studies and ground water development. The Committee submitted its recommendations in the year 1984 and suggested a methodology (GEC-1984) for assessment of dynamic ground water resources. As per the recommendations of the GEC-1984 the State Governments were advised to constitute Working Groups for

assessment of ground water potentials. The Working Groups were headed by Irrigation Secretaries-In charge of Ground Water Developments and included Heads of Ground Water Department, State Agriculture Departments, representatives from Agriculture Universities and NABARD. Director, CGWB was the convener of the group. The base year for the computation of the resource mostly varied between 1991 and 1993 and a National report on Ground Water Resources of India was brought out in 1995 by compiling the data of all the States and Union Territories of the country. As per the report, the total Replenishable Ground Water in India was estimated to be about 432 billion cubic meter. The ground water resource available for irrigation purpose was about 361 billion cubic meter. The Net Ground Water Draft from Irrigation uses was around 115 billion cubic meters and the level of development was 32%. The volumetric resource was converted in terms of area and the Utilizable Irrigation Potential from ground water of the country was worked out to be 64 million hectare (Central Ground Water Board, 1995).

Increasing thrust on ground water and changed scenario of data acquisition led the Government of India to form another Committee in 1995 to review the existing methodology for ground water resource assessment and to suggest revisions if necessary. The Committee submitted its report in 1997 wherein a revised and elaborate methodology for resource assessment has been suggested, more commonly called as GEC-1997. While estimating the ground water resources in the hard rock terrains some limitations have been observed. To address these limitations another Committee on Ground Water Estimation Methodology in Hard Rock Terrain was formed in 2001 to review the existing methodology for resource estimation in hard rock terrains. The Committee made certain suggestions on the criteria for categorization of blocks to be adopted for the entire country irrespective of the terrain conditions. Based on GEC-1997, the dynamic ground water resources of India were estimated for the entire country with 2004 and then again with 2009 as base year. In 2004, the Annual Replenishable Ground Water Resources is 433 billion cubic meter (bcm). Keeping an allocation for Natural Discharge, the

net annual ground water availability was 399 bcm and the annual ground water draft for all uses was of the order of 231bcm. The overall stage of ground water development for the entire country was 58%. In 2009, the annual replenishable ground water resources are 431 bcm. The net annual ground water availability of the country has been assessed as 396 bcm after deducting 35 bcm for natural discharge. The annual ground water draft of the country as on March, 2009 was 243 bcm. The largest use is the irrigation sector (about 91%). The stage of ground water development for the entire country has been computed as 61%. The utilization pattern is however uneven across the country resulting in ground water stressed conditions in some parts of the country while in other areas, ground water utilization have been sub-optimal. In 2009, assessments were carried out in 5842 assessment units (blocks/mandals/taluks) in the country, out of which around 14% (802 units) were categorized as Over-exploited. The critical units were 3% (169 units), semi-critical 9% (523 units) and safe were 73% (4277 units). The remaining units 71 (less than 1%) were having totally saline ground water.

1.2 Ground water management initiatives

The findings of the ground water resources assessment guide the planners and stakeholders to take appropriate management measures for optimal utilization and sustainability of the resource. Ground water resources in the country are withdrawn by approximately 20 million wells (World Bank, 2010) mostly through private initiatives. It is a common-pool resource, used by millions of farmers across the country. Management of this resource is hence done both through Govt. and private initiatives. The areas where future scope for ground water development exists, Banks and NABARD provide loans for sinking of wells and installation of pump sets. Various Govt. schemes also exist to sponsor well construction in Safe blocks. On the other hand, measures are being taken to control further deterioration of the resources in Over-exploited and Critical blocks. Ground water management practices like water conservation and water harvesting measures, Farmer's Participatory Action Research Programme (FPARP), efficient water use practices with

community participation, restrictions on institutional funding for ground water irrigation in Over-exploited and Critical blocks are some such examples. Community management of ground water resources has been quite effective in certain places in the country. Success stories on effective community participation on water conservation measures such as community management and water conservation measures in Aravalli hills of Rajasthan, Ralegaon Sidhi, Hiware Bazar and World Bank sponsored Projects in rural Maharashtra, Andhra Pradesh Farmer-Managed Groundwater Systems Project (APFAMGS), Water Conservation measures and efficient water use practices in Gujarat are well known. There are several such efforts across the length and breadth of the country. All these initiatives have helped in improving the ground water situation in localized areas in certain parts of the country.

1.3 Re-assessment of Ground Water Resources (2010-11)

In order to assess prevailing ground water scenario of the country, re-assessment of ground water resources was felt necessary after two year interval. It would also indicate the impact of the on-going ground water management practices on the ground water regime. In 2012, Ministry of Water Resources constituted a Central Level Expert Group (CLEG) for over-all supervision of the re-assessment of ground water resources in the entire country. The terms of reference of the Group include supervision of assessment of annual replenishable ground water resources and the status of utilization for the reference year 2010-11. The copy of the Government Resolution is given as **Appendix A**. Ministry of Water Resources requested all the State Governments to constitute State Level Committees for overall supervision of assessment of the groundwater resources at the State level (**Appendix B**). The ground water resources assessment for 2010-11 at the States have been carried out jointly by State Ground Water Departments and Central Ground Water Board under the supervision and final approval of the State level Committees (**Appendix C**). Central Level Expert Group provided the technical guidance in this regard. Based on the assessment figures

provided by the State Government, the national report has been compiled under the supervision of the CELG for overall reassessment of the ground water resources of the country. The meetings of the CLEG were held during 2012-13 to finalize the report on dynamic ground water resources of India (as on 31 March 2011) and was approved in the 2nd meeting of CLEG (**Appendix D**). List of the members and special invitees who have attended the meetings of the Central Level Expert Group is given in **Appendix E**.

CHAPTER 2

GROUND WATER RESOURCES ESTIMATION METHODOLOGY

The ground water resources assessment for 2008-09 were carried out based on the guidelines of Ministry of Water Resources which broadly follows the methodology recommended by Ground Water Resources Estimation Committee, 1997. However, there have been few modifications in the MOWR guidelines namely in the criteria of Categorization of Assessment units and future Allocation for irrigation and domestic & industrial sector. Further MOWR suggested a protocol for assessment of ground water resources. The salient features of the methodology are enumerated in the following paragraphs.

Ground water resources are estimated assessment unit wise. The assessment unit is watershed in the states occupied predominantly with hard rocks. This is because the ground water balance equations recommended in GEC-1997 can be better applied in the assessment units with hydrologic/hydrogeological boundaries. However, in the states covered predominantly with alluvium and/ or soft rocks, administrative blocks are chosen as assessment unit since in alluvial areas it is difficult to identify watershed considering the possibility of trans-boundary aquifer system. Within the assessment areas, the hilly areas (slope greater than 20%) are to be excluded since these are not likely to contribute to ground water recharge. The assessment units are to be divided into command and non-command areas for the purpose of computation of ground water resources. The ground water resources in the poor quality (saline) areas are to be computed separately (Ministry of Water Resources, 1997).

The ground water recharge is estimated season-wise both for monsoon season and non-monsoon season separately. The following recharge and discharge components are assessed in the resource assessment - recharge from rainfall, recharge from canal, return flow from irrigation, recharge from

tanks & ponds and recharge from water conservations structures and discharge through ground water draft.

2.1 Assessment of ground water draft

Ground water draft is estimated seasonally. The most commonly used method for computation of irrigation draft is – number of abstraction structures multiplied by the unit seasonal draft. Alternative methods like area irrigated by ground water and the associated crop water requirements are also recommended for assessment of ground water draft for irrigation. Ground water draft for Domestic & Industrial needs is computed using unit draft method and based on consumptive use pattern of the population.

2.2 Assessment of ground water recharge from other sources

Ground water recharge due to return flow from irrigation, seepage from canals, recharge from tanks and ponds and recharge from water conservation structures are to be estimated separately for both monsoon and non-monsoon seasons based on the recommended norms as given in **Table 1**.

Table -1 Recommended norms for the Assessment of Recharges from Other Sources

Parameters	Sources of Recharge	Range of Parameters
Canal seepage factor	Unlined canals	15 to 30 ham/day/million sq.m. of wetted area
	Lined canals & canals in hard rock terrain	20% of above value suggested for unlined canals
Return flow factor	Surface water Irrigation	0.10 – 0.50
	Ground water Irrigation	0.05 – 0.45
Seepage from tanks and ponds	1.4 mm/day over the average water spread area	
Water conservation structures	50% of the Gross Storage. Out of this, 50% is during monsoon season and the remaining 50% during non-monsoon season	

(Source: Ministry of Water Resources, 1997)

2.3 Assessment of ground water recharge from rainfall

Ground water recharge from rainfall is estimated for monsoon and non-monsoon seasons separately.

Rainfall recharge during monsoon season is estimated using two methods – Water level fluctuation Method and Rainfall Infiltration Factor Method.

2.3.1 Water level Fluctuation (WLF) Method

Under this method the change in storage will be computed by multiplying water level fluctuation between pre and post monsoon seasons with the area of assessment and specific yield.

$$\text{Change in Storage} = \Delta S = h * S_y * A \quad \text{.....(i)}$$

Where,

h = rise in water level due to monsoon (fluctuation between pre-monsoon and post-monsoon water level), A = area for computation of recharge, S_y = specific yield.

The Specific yield of a soil or rock is the ratio of the volume of water that, after saturation, can be drained by gravity to its own volume (Todd & Mays, 2005). The Specific yield data were either arrived through field studies, including long-duration pumping tests and dry season groundwater balance (in hard-rock areas) or adopted from the norms recommended by GEC-1997, which were derived from the various water-balance studies carried out by CGWB, SGWDs and academic/research institutions. The range of specific yield considered for different formations are given in the **Table 2**.

Table - 2 Specific Yields for Different Formations

Formation		Range of Specific Yield
Unconsolidated formations	Alluvium	0.04 to 0.22
Semi-consolidated formations	Sedimentary rocks	0.01 to 0.15
Consolidated formations	Crystalline and other hard rocks	0.002 to 0.04

The change in storage calculated from the above relation is the resultant of the recharge from rainfall and other sources during the monsoon period and the gross ground water draft during monsoon season. In order to segregate the rainfall recharge during monsoon season, the following equation is used –

$$R_{rf} = h \times S_y \times A + DG - R_c - R_{sw} - R_t - R_{gw} - R_{wc} \quad \dots\dots(ii)$$

Where,

DG = Gross ground water draft for all uses during monsoon season

R_c = recharge due to seepage from canals during monsoon season

R_{sw} = recharge from surface water irrigation during monsoon season

R_t = recharge from tanks and ponds during monsoon season

R_{gw} = recharge from ground water irrigation during monsoon season

R_{wc} = recharge from water conservation structures during monsoon season

The rainfall recharge thus calculated is normalized for the normal monsoon season rainfall.

2.3.2 Rainfall Infiltration Factor (RIF) Method

The other method for assessment of rainfall recharge is using Rainfall infiltration factor. The recharge from rainfall is to be estimated as given below

$$R_{rf} = f \times A \times \text{normal monsoon rainfall} \quad \dots\dots(iii)$$

Where;

f = rainfall infiltration factor

A = area

The same Rainfall Infiltration Factor should be used for computation of recharge due to rainfall during monsoon and non monsoon seasons.

The norms adopted for computation of recharge from rainfall is given in **Table – 3**.

Table – 3 Rainfall Infiltration Factor for different formations

Formation		Range of Rainfall Infiltration Factor
Unconsolidated formations –	Alluvium	0.08 to 0.25
Semi-consolidated formations	Sedimentary rocks	0.03 to 0.14
Consolidated formations	Crystallines and other hard rocks	0.01 to 0.12

(Source: Ministry of Water Resources, 1997)

The rainfall recharge computed by WLF method is to be compared with recharge computed by RIF method. In case the difference between the two sets of data are more than 20%, then rationalized RIF figure is to be considered, otherwise monsoon recharge using WLF method is to be considered. Whenever the percent difference is less than - 20%, 80 % of the recharge computed by RIF method is to be used and wherever, the percent difference is more than + 20 %, 120 % of recharge computed by RIF method is to be taken.

2.4 Ground water Recharge during Monsoon Season

The total recharge in monsoon season is the sum of the normalized rainfall recharge and the recharge from other sources as expressed in the following equation –

$$R(\text{normal}) = R_{rf}(\text{normal}) + R_c + R_{sw} + R_t + R_{gw} + R_{wc} \quad \dots\dots\dots(\text{iv})$$

Where,

$R(\text{normal})$ = Total recharge during monsoon season

$R_{rf}(\text{normal})$ = Rainfall recharge during monsoon season for normal monsoon season rainfall

2.5 Ground water Recharge during non-Monsoon Season

Similar expression as given in equation (iv) above is used for recharge during non-monsoon season wherein all the recharge components including rainfall

recharge and recharge from other sources during non-monsoon season are computed. Only difference is that rainfall recharge during non-monsoon is computed using RIF method only. If the rainfall during non-monsoon period is less than 10% of the annual rainfall, the recharge due to rainfall is taken as zero. The total recharge during non monsoon is the sum of recharge from rainfall and recharge from other sources.

2.6 Annual Replenishable Ground Water Resources

The Annual Replenishable Ground Water Resources of the area is the sum of recharge during monsoon and non monsoon seasons. An allowance is kept for natural discharge during non monsoon season by deducting 5% of Annual Replenishable Ground Water Resource, wherever WLF method is employed to compute rainfall recharge during monsoon season and 10% if RIF method is used.

2.7 Net Annual Ground Water Availability

The Net annual ground water availability is the available resource after deducting the natural discharges from the Annual Replenishable Ground Water Resource and is expressed as:-

Net Annual Ground Water Availability = Annual Replenishable Ground Water Resource – Natural Discharge during non monsoon season (v)

2.8 Future Utilization of Ground Water Resources

The projected demand for domestic and industrial water supply is kept based on projected population for the year 2025 and present dependency on ground water. The ground water available for future irrigation is obtained by deducting the sum of projected demand for Domestic and Industrial use and existing gross irrigation draft from the Net Annual Ground Water Availability. In order to rationalize the projected demand of ground water resources in over-exploited areas, following procedure is adopted:

- Ø Case I, when $GW_{av} \geq D_{gi} + A_{lId}$
In such cases projected demand for future domestic and industrial uses = A_{lId}
- Ø Case II, when $GW_{av} < D_{gi} + A_{lId}$

In such cases, projected demand for future domestic and industrial uses = (GWav – Dgi) or Dgd, whichever is more.

Where,

GWav = Net Annual Ground Water Availability

Dgi = Existing Ground Water draft for Irrigation

Dgd = Existing Ground Water draft for Domestic use

Dg = Existing Ground water draft for all uses

Alld = Computed value of allocation for domestic use

(Based on projected population, fractional load on ground water and per capita requirement)

2.9 Stage of Ground Water Development

The stage of Ground water Development is to be computed as given below,

$$\text{Stage Of Development} = \frac{\text{Existing Gross Draft For All Uses}}{\text{Net Annual Groundwater Availability}} \times 100 \dots\dots\dots(vi)$$

2.10 Categorization of Assessment Units

The assessment units are to be categorized for ground water development based on two criteria – a) stage of ground water development, and b) long-term trend of pre and post monsoon water levels. The long term ground water level trend is to be computed generally for a period of 10 years. The significant rate of water level decline has been taken between 10 and 20 cm per year depending upon the local hydrogeological conditions. There are four categories, namely – ‘Safe’, ‘Semi-critical’, ‘Critical’ and ‘Over-exploited’ areas. The criteria for categorization are given below.

Table - 4 Criteria for Categorization of Assessment Units

Stage of Ground Water Development	Significant Long Term Water level Decline trend		Category
	Pre-Monsoon	Post-Monsoon	
<=90%	No	No	Safe
>70% and <=100%	No	Yes	Semi-Critical
>70% and <=100%	Yes	No	Semi-Critical
>90% and <=100%	Yes	Yes	Critical
>100%	No	Yes	Over-Exploited
>100%	Yes	No	Over-Exploited
>100%	Yes	Yes	Over-Exploited

Apart from the four categories mentioned above, blocks where the entire assessment area is having poor quality ground water are demarcated as Saline blocks.

A sample calculation of one assessment unit is presented as **Appendix F** to illustrate the methodology for assessment of replenishable ground water resources.

The State Governments broadly followed the methodology outlined above while carrying out the computation of ground water resources. However, at some places, the detailed steps, norms and criteria for categorization have been modified by the States to match the prevailing ground water conditions in the field.

CHAPTER 3

RAINFALL OF INDIA

Rainfall is the main source of ground water recharge. India receives about 120 cm of rain in a year. However distribution of rainfall has a wide variation both in space and time. Most of rainfall (about 75%) occurs during a short span of four Monsoon months (June to September) resulting into eight relatively dry months. Similarly the meteorological Subdivisions like Coastal Karnataka, Konkan and Goa, North east India receive more than 250 cm of rainfall annually whereas West Rajasthan gets only about 30cm (IMD, 2011).

3.1 Rainfall Pattern

Rain gauge stations are established and maintained by different departments and Undertakings of Central and State governments and also by private parties to cater their particular data need. India Meteorological Department (IMD) has 559 observatories (both departmental and part time) while amongst non-Departmental Rain gauge Stations, 3540 are reporting and 5039 are non-reporting.

Though the period of seasons varies from place to place, for Climatological purposes especially for rainfall, year at all the places is uniformly divided into 4 parts, called seasons. The seasons are: Winter (January and February), Pre monsoon (March to May), South West Monsoon (June to September) and Post Monsoon season (October to December).

For the purpose of compiling the rainfall data and draw the inferences India is divided into 36 meteorological homogeneous regions.

The long term average rainfall indicates that most part of India receives rainfall mainly during SW Monsoon season. However main Rainfall season in Tamil Nadu is Post Monsoon season. Jammu and Kashmir, Himachal Pradesh and Uttarakhand receive significant rainfall in all 4 seasons.

Normal Annual rainfall (1941 to 1990) for each Meteorological Sub Division with Coefficient of Variation (measure of dispersion of a probability distribution) is shown in **Figure-1**. Maximum variation is observed in NW India while minimum variation is observed in coastal regions and Assam-Meghalaya area. The map of normal annual rainfall (**Figure-2**) shows that highest rainfall of more than 250 cm is received in the North Eastern States and along West Coast in the Konkan region. The major part of country including Northern, Central and Eastern India receives rainfall between 75 and 150 cm. A portion of Peninsular India including parts of Western Maharashtra and Karnataka receives less rainfall up to 75 cm. Western part of the country including Rajasthan and Gujarat receives the lowest rainfall in the country, at places even less than 15 cm in an year i.e. in western part of Rajasthan.

Rainy day is defined as a day when rainfall recorded is at least 2.5 mm. Number of rainy days in different parts is shown in **Figure-3**. For given amount of rainfall, more number of rainy days is favourable for ground water recharge. Annual number of rainy days is less than 10 in parts of West Rajasthan, Gujarat and Kashmir, gradually increasing towards east and more than 100 days in North East portion of the country.

Perusal of Monsoon rainfall maps of 2009 to 2011 (**Figure-4**) indicate that overall rainfall pattern has been normal during intervening period between two successive ground water assessment viz. 2009 and 2011. In 2008, major part of the country received normal monsoon rainfall, except for Punjab and Orissa, where excess monsoon rainfall was recorded and Western Madhya Pradesh, Vidharbha and Kerala which received deficient monsoon rainfall. In 2009, major part of the country received deficient rainfall where Arunachal Pradesh, Gangetic West Bengal, Madhya Maharashtra Konkan, Goa, Coastal Karnataka Kerala Tamilnadu, Puducherry and Rayalseema received normal rainfall. Ondly some part of country i.e. NI&SI Karnataka, Saurashtra Kutch, Daman & Diu received excess rainfall. In 2010 most part of country has received excessive rainfall like J&K, Uttarachal, Haryana, Chandigarh, West

Rajasthan, Saurashtra, Kutch, diu, Marathwada, Madhya Maharashtra, Vidarbha, Telegana, Rayalseema, Tamilnadu and Puducherry whereas Assam, Meghalaya, Bihar, East Uttar Pradesh, Jharkhand and Gangetic West Bengal has got deficient rainfall. In 2011 West & East Rajasthan, West MP, Saurashtra, Kutch, diu, Konkan, Goa and coastal Karnataka received excessive rainfall whereas Arunachal Pradesh, Assam and NMMT has got deficient rainfall. Remaining part of country received normal rainfall.

Analysis of average monsoon rainfall data for the period 2009 to 2011, shows that in major parts of the country, 31 met-subdivisions out of 36, the average rainfall for the three year period (2009 to 11) is within the normal range (**Table No.5**). However, West Rajasthan and Saurashtra and Kutch recorded excess rainfall. The N.E. regions like NMMT, Assam & Meghalaya and the met-subdivisions East U.P. of North West India recorded deficient rainfall. For the same period the average rainfall for the country as a whole was 6% less than its LPA i.e. the rainfall for the country as a whole was 836.6 mm against its normal value of 890.8 mm. The annual average rainfall pattern for the same period 2009 to 2011, was almost same as that of monsoon rainfall scenario. However, three met-subdivisions Jharkhand, Punjab and Himachal Pradesh, which were normal during the monsoon season recorded less rainfall on annual scale and remained deficient. The annual average rainfall for the country as a whole was recorded 8% less than its LPA i.e. the annual rainfall for the country as a whole was 1094.3 mm against its normal value of 1193.1 mm. It is noticed that met-subdivisions West Rajasthan and Saurashtra & Kutch recorded excess rainfall during SW monsoon as well as annual statistics for the whole period of 2009 to 2011.

3.2 Trends in Rainfall

One of the most significant consequences of global warming due to increase in greenhouse gases would be an increase in magnitude and frequency of extreme precipitation events. National Climate Centre of India meteorological Department has studied Trends in the extreme rainfall indices

for the period 1901-2000 for 100 stations over India and published same in National Climate Centre Research Report No: 2/2006. Report depicts that Rainfall changes expected for 36 Meteorological sub Divisions for 4 seasons and Annual. **Figure-5** shows Increase/Decrease in rainfall in mm in 100 year in each of 36 subdivisions for annual.

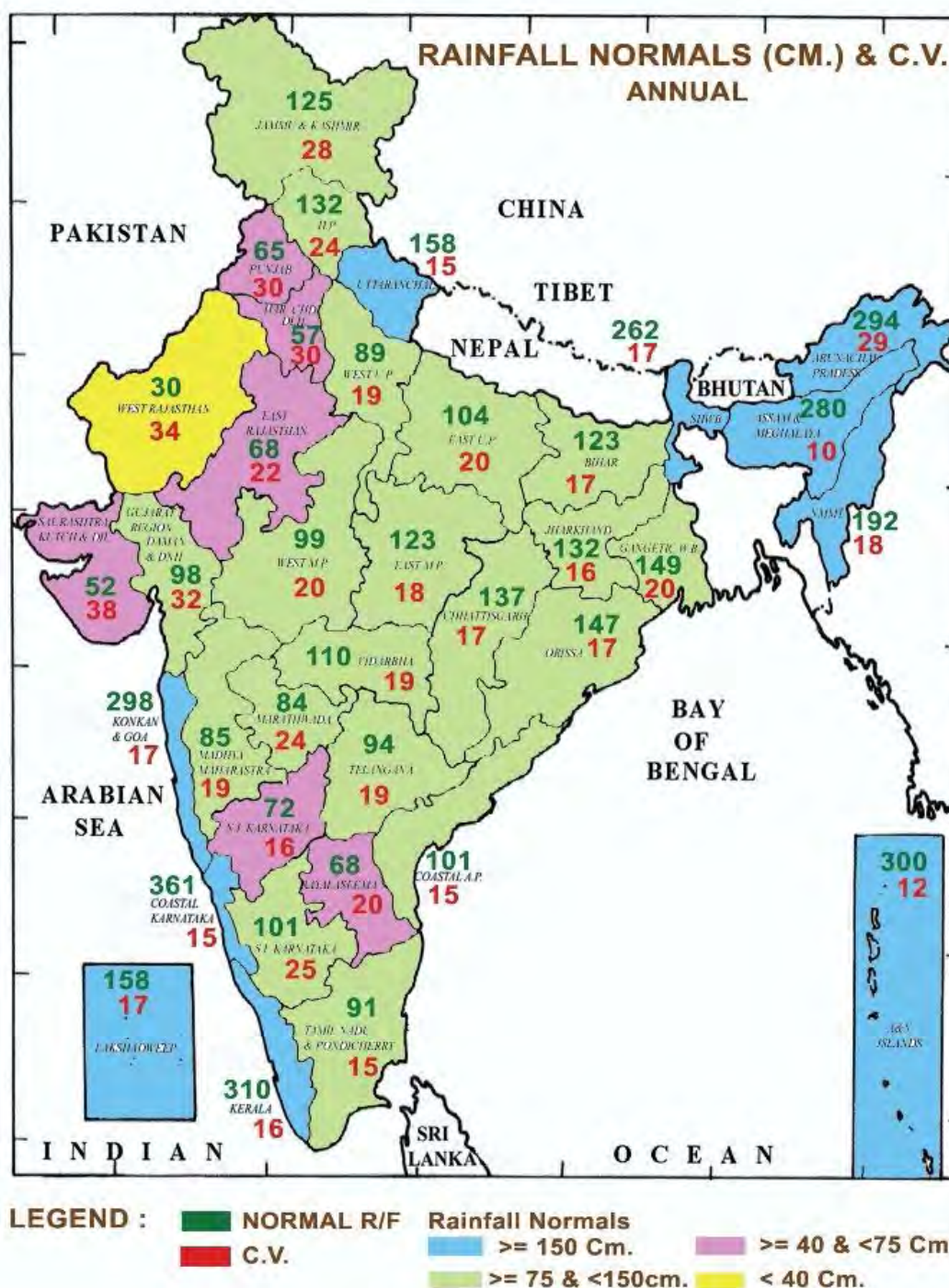
From the Figure, it may be seen that Significant increasing trend is observed in the annual scale for the sub-divisions Konkan & Goa, Madhya Maharashtra, North Interior Karnataka, Rayalaseema, coastal Andhra Pradesh, Gangetic West Bengal, Assam & Meghalaya and Jammu & Kashmir. On the other hand significant decrease in rainfall is observed for the sub-divisions Chhattisgarh, Jharkhand and Kerala.

Table-5 Subdivision-wise Monsoon Averages Rainfall(mm) (2009-2011)(IMD)

S.	METEOROLOGICAL	2009-2011 (JUNE-SEPT.)			
NO.	SUBDIVISIONS	AVERAGE	NORMAL	%DEP.	CAT
1.	A & N Island	1906.2	1710.3	11%	N
2.	Arunachal Pradesh	1488.1	1770.8	-16%	N
3.	Assam & Meghalaya	1365.6	1876.6	-27%	D
4.	N M M T	1107.5	1378.8	-20%	D
5.	Shwb & Sikkim	1863.5	1962.4	-5%	N
6.	Gangetic West Bengal	1050.2	1148.3	-9%	N
7.	Orissa	1089.1	1161.4	-6%	N
8.	Jharkhand	882.6	1089.6	-19%	N
9.	Bihar	868.4	1030.4	-16%	N
10.	East U.P.	696.3	906.9	-23%	D
11.	West U.P.	646.2	771.1	-16%	N
12.	Uttarakhand	1346.2	1220.1	10%	N
13.	Har. Chd & Delhi	417.2	467.9	-11%	N
14.	Punjab	416.5	496.5	-16%	N
15.	Himachal Pradesh	705.1	791.0	-11%	N
16.	Jammu & Kashmir	510.9	524.1	-3%	N
17.	West Rajasthan	331.3	262.8	26%	E
18.	East Rajasthan	639.6	623.2	3%	N
19.	West Madhya Pradesh	822.3	894.6	-8%	N
20.	East Madhya Pradesh	952.5	1078.7	-12%	N
21.	Gujarat Region	842.0	915.0	-8%	N
22.	Saurashtra & Kutch	779.5	482.1	62%	E
23.	Konkan & Goa	3156.7	2838.6	11%	N
24.	Madhya Maharashtra	761.2	710.2	7%	N
25.	Marathwada	689.0	699.4	-1%	N
26.	Vidarbha	924.7	968.6	-5%	N
27.	Chhattisgarh	1017.1	1185.4	-14%	N
28.	Coastal Andhra Pradesh	600.9	577.2	4%	N
29.	Telangana	722.6	763.0	-5%	N
30.	Rayalaseema	429.4	386.7	11%	N
31.	Tamilnadu & Pondicherry.	329.9	315.5	5%	N
32.	Coastal Karnataka	3409.3	3143.9	8%	N
33.	N. I. Karnataka	557.2	496.0	12%	N
34.	S. I. Karnataka	734.4	663.8	11%	N
35.	Kerala	2035.3	2107.4	-3%	N
36.	Lakshadweep	1055.6	989.6	7%	N
Country as a whole		836.6	890.8	-6%	N

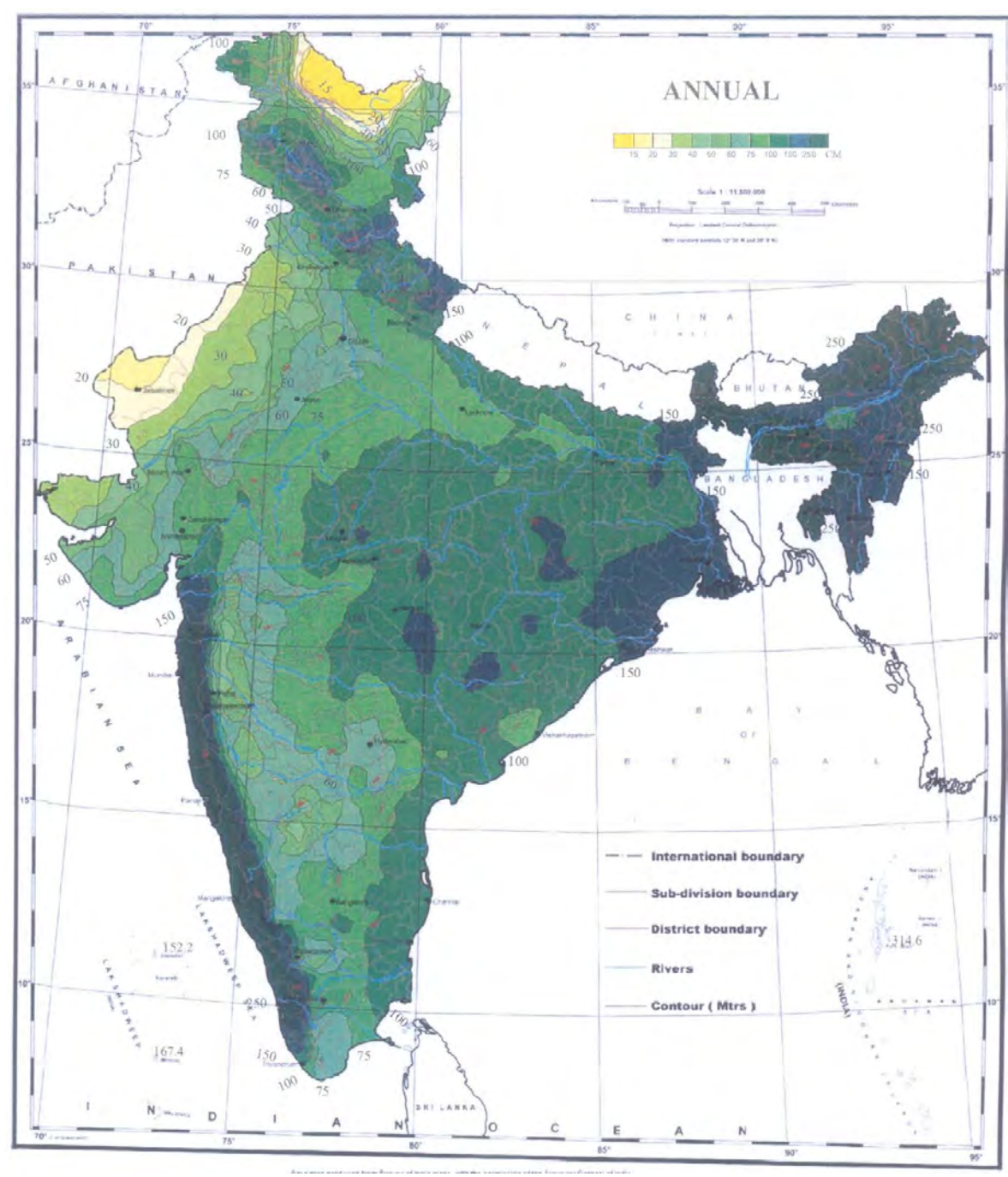
Excess: 2, Normal: 31, Deficient: 3, Scanty: 0, No Rain: 0

Figure-1



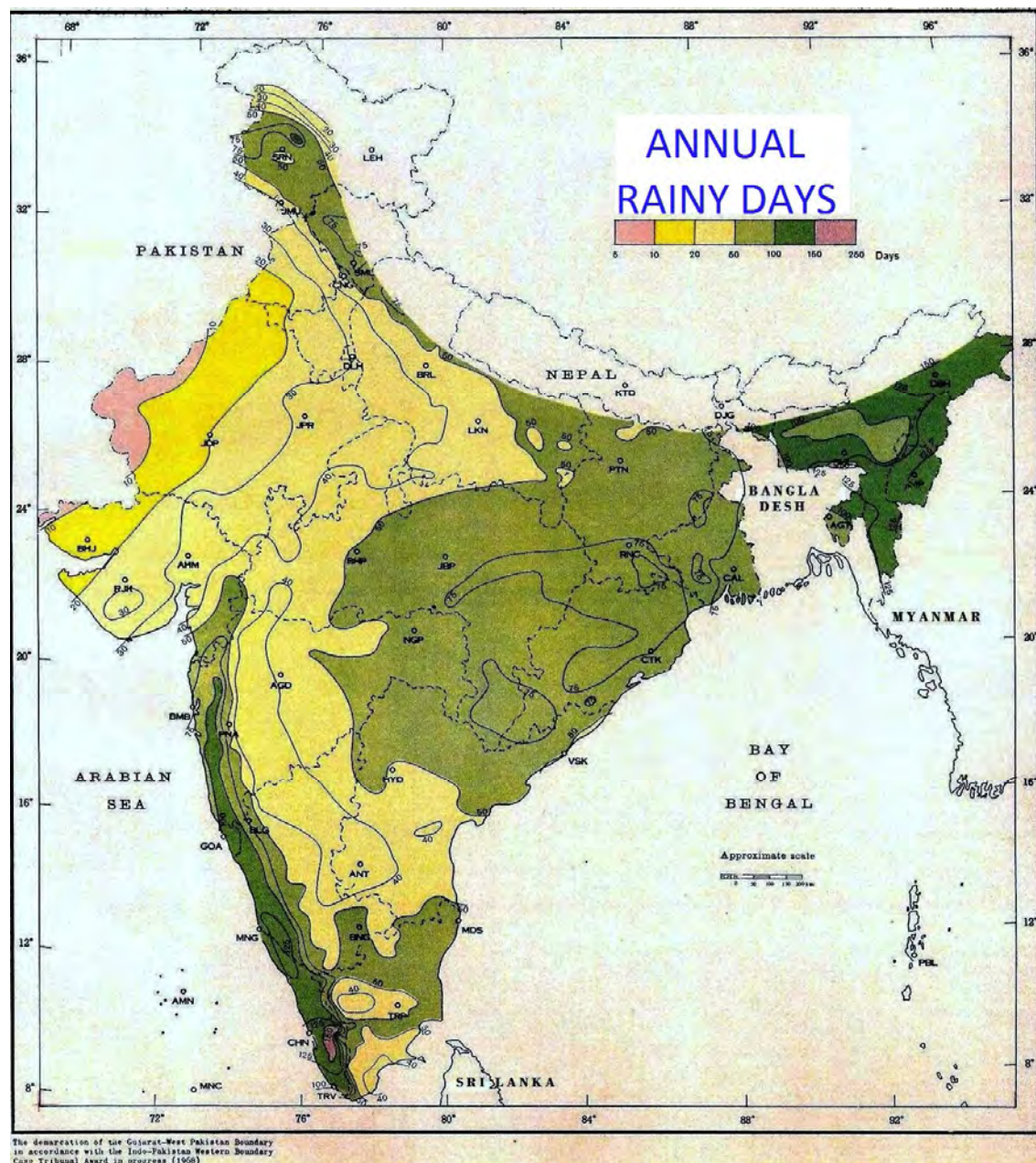
Source: India Meteorological Department

Figure-2 Normal Annual Rainfall



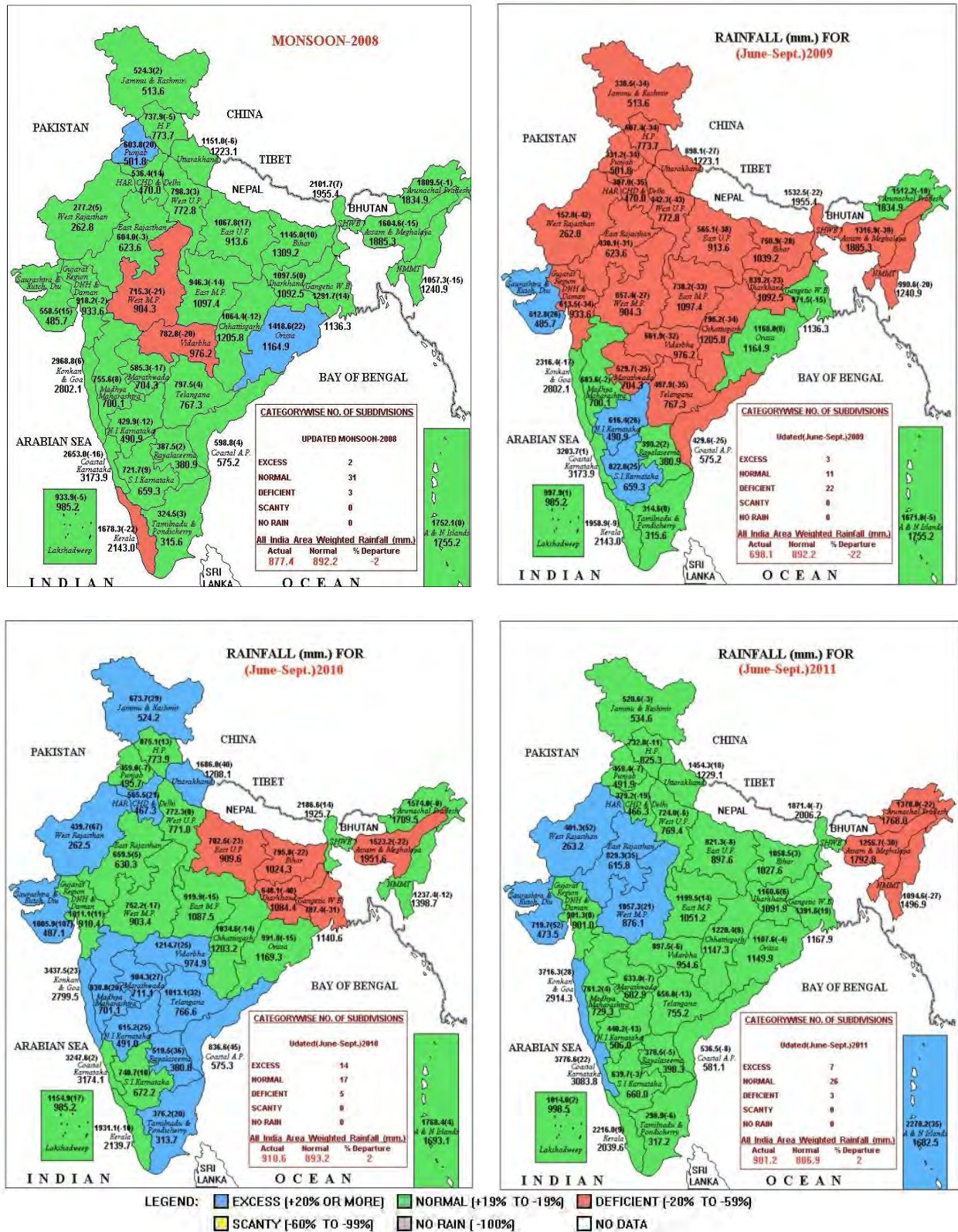
Source: India Meteorological Department

Figure-3 Annual Normal Rainy Days



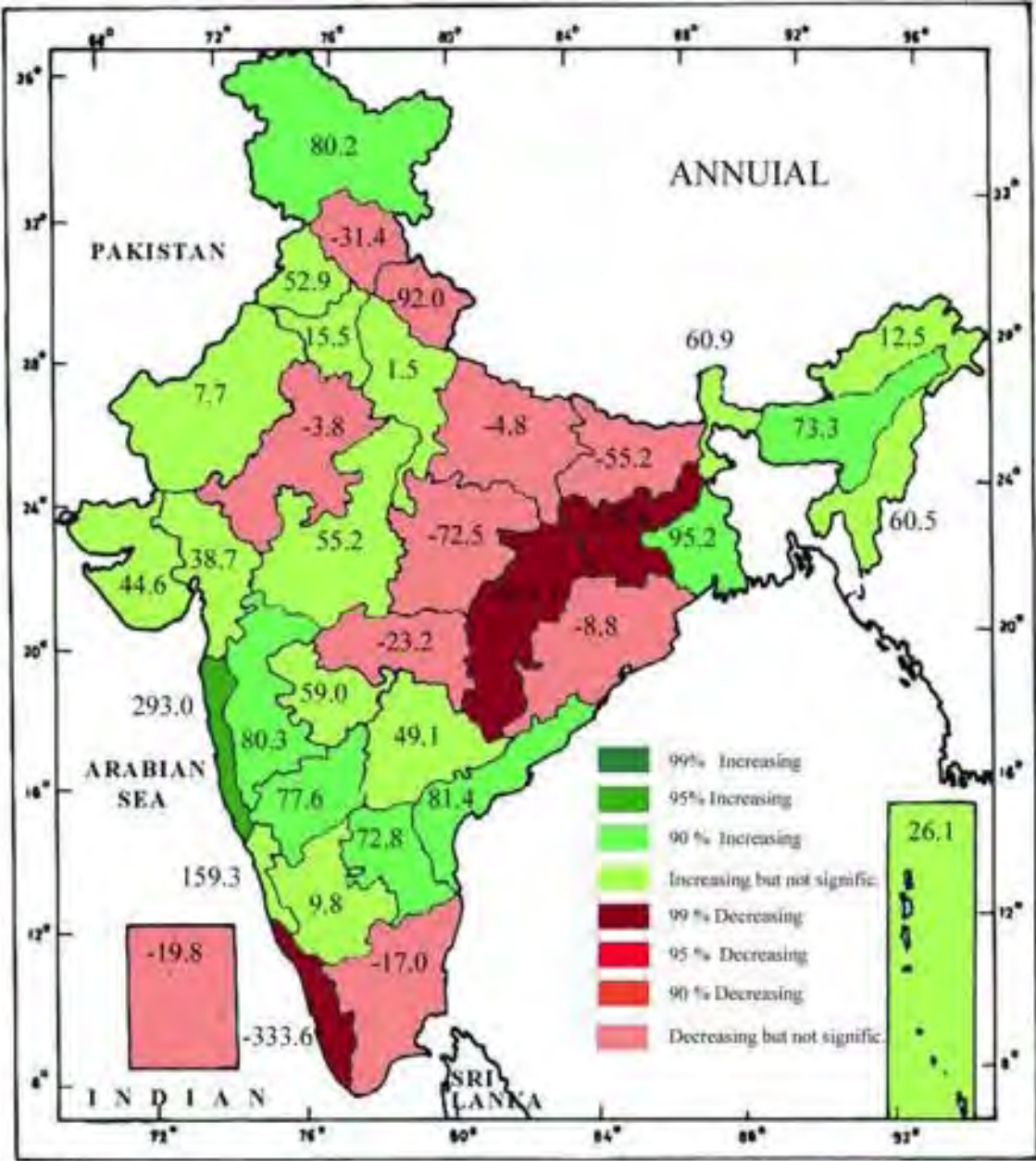
Source: India Meteorological Department

Figure-4 Year-wise monsoon rainfall distribution (2008-2011)



[Source: India Meteorological Department]

Figure-5. Increase/Decrease in annual rainfall (in mm) in 100 years period for each of 36 subdivisions of India



[Source: India Meteorological Department]

3.3 Drought analysis

Drought is a short term extreme climatic events. About 68% of the country is prone to drought in varying degrees.

There are three conditions that are generally referred to as drought:

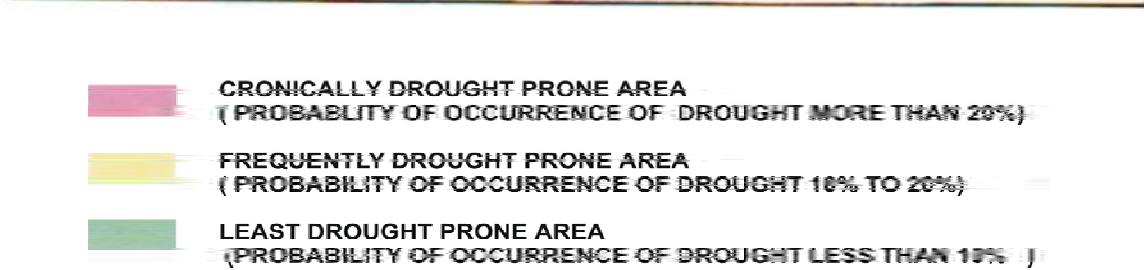
Meteorological drought:- This type of drought all about the weather and occurs when there is a prolonged period of below average precipitation, which creates a natural shortage of available water. Country as whole has received normal rainfall.

Agricultural drought:- This type of drought occurs when there isn't enough moisture to support average crop production on farms or average grass production on range land. Although agricultural drought often occurs during dry, hot periods of low precipitation, it can also occur during periods of average precipitation when soil conditions or agricultural techniques require extra water.

Hydrological drought:- This type of drought occurs when water reserves in aquifers, lakes and reservoirs fall below an established statistical average. Again, hydrological drought can happen even during times of average or above average precipitation, if human demand for water is high and increased usage has lowered the water reserves.

Drought classification of all India rainfall time series indicates that 11 years viz. 1901, 1904, 1907, 1913, 1920, 1941, 1966, 1968, 1974, 1986 & 2004 were under moderate drought category, 6 years (1905, 1911, 1951, 1965, 1982 & 2002) were under severe drought category and 5 years (1918, 1972, 1979 & 1987, 2009) were under extreme drought category. Based on frequency of drought and probabilities for 36 sub-divisions of 130 years data, highest frequency is observed in West Rajasthan and Saurashtra & Kutch with 31 cases. The adjoining Gujarat Region which mostly belongs to semi-arid also experienced high incidence of drought. Jammu Kashmir regions which mostly received rains from extra-tropical systems in most-monsoon and winter season also faces numerous incidences of drought. Other areas recording significantly large incidence of drought are Haryana, Delhi, Chandigarh, Punjab,

Himachal Pradesh and East Rajasthan in northwest India and Rayalseema in southern Peninsula. The lowest number of droughts have been observed, for obvious reasons, in Arunachal Pradesh, Assam, Meghalaya, Orissa, Gangetic West Bengal and Jharkhand and coastal area of Karnataka also experienced significant less frequency of droughts (**Fig.6**). Konkan and Kerala occasionally experienced more drought.



CHAPTER 4

HYDROGEOLOGICAL SETUP OF THE COUNTRY

Two groups of rock formations have been identified depending on characteristically different hydraulic parameters of these formations viz. Porous Formation and Fissured Formation. Studies carried out over the years have revealed that aquifer groups in alluvial / soft rocks even transcend the surface basin boundaries

India is occupied by a variety of hard and fissured formations, including crystalline, trappean basalt and consolidated sedimentaries (including carbonate rocks), with patches of semi-consolidated sediments in narrow intra-cratonic basins. Apart from this, the central part of the country is occupied by alluvial formation stretching from Rajasthan in the west to Brahmaputra valley in the east. Rugged topography, compact and fissured nature of the rock formations, combine to give rise to discontinuous aquifers, with moderate to poor yield potentials. The near surface weathered mantle forms an important aquifer in case of hard rocks. In hard rock terrains, deep weathered pediments, lowlands, valley fills and abandoned river channels, generally have adequate thickness of porous material, to act as repositories of groundwater.

4.1 Aquifer Systems of India

The various rock formations with different hydrogeological characteristics act as distinct aquifer systems of various dimensions. The aquifer systems of India can be broadly categorized in to 14 Principal Groups .A brief description of the Principal Aquifer Systems is given below. The principal Aquifer systems identified by CGWB are given in **Fig 7**.

Alluvial aquifers

The Quaternary rocks comprising Recent Alluvium, Older Alluvium, Aeolian Alluvium (Silt/ Sand) and Coastal Alluvium of Bay of Bengal are by and large important unconsolidated formations comprising major alluvial aquifers.

These sediments are essentially composed of clays, silts, sands, pebbles, Kankar etc.

. These are by far the most significant ground water reservoirs for large scale and extensive development. The hydrogeological environment and ground water regime conditions in the Indo-Ganga-Brahmaputra basin indicate the existence of potential aquifers having enormous fresh ground water reserves. Bestowed with high incidence of rainfall and covered by a thick pile of porous sediments, these ground water reservoirs get replenished every year and are being used heavily. In these areas, in addition to the Annual Replenishable Ground Water Resources available in the zone of Water Level Fluctuation (Dynamic Ground Water Resource), there exists a huge ground water reserve in the deeper passive recharge zone below the zone of fluctuation as well as in the deeper confined aquifers which is nearly unexplored. The coastal aquifers show wide variation in the water quality, both laterally and vertically, thus imposing quality constraints for groundwater development.

Laterite

Laterites are formed from the leaching (chemical weathering) of parent sedimentary rocks (sandstones, clays, limestones); metamorphic rocks (schists, gneisses, migmatites) and igneous rocks (granites, basalts, gabbros, peridotites). It is rich in iron and aluminium, formed in hot and wet tropical areas. Laterites are the most wide spread and extensively developed aquifer especially in the peninsular states of India. Laterite forms potential aquifers along valleys and topographic lows where the thickness of the saturated zone is more and can sustain large diameter open wells for domestic and irrigation use.

Sand stone, shale aquifers

The sand stone and shale aquifers generally belong to the group of rocks ranging in age from Carboniferous to Mio-Pliocene. These are mainly composed of shale and sandstone. The terrestrial freshwater deposits

belonging to Gondwana System and the Tertiary deposits along the west and east coast of the peninsular region are included under this category. The Gondwana sandstones form highly potential aquifers, locally. Elsewhere, they have moderate potential and in places they yield meagre supplies. The Gondwanas, Lathis, Tipams, Cuddalore sandstones and their equivalents are the most extensive productive aquifers in this category.

Lime stone aquifer

The consolidated sedimentary rocks include carbonate rocks such as limestones, dolomite and marble. Among the carbonate rocks, limestones have the greatest distribution. In the carbonate rocks the principal aquifers are the fractured zone and solution cavities. Consolidated sedimentary rocks of Cuddapah and Vindhyan subgroups and their equivalents consist of limestones/dolomites apart from other major litho-units such as conglomerates, sandstones, shales, slates and quartzites.

Basalt aquifers

Basalt is the basic volcanic rock which forms alternate layers of compact and vesicular beds of lava flows as seen in the Deccan trap area. The groundwater occurrence in the basalts are controlled by nature and extent of weathering, presence of vesicles and lava tubes, thickness, number of flows and the nature of inter-trappean layers. The basalts have usually medium to low permeability. Groundwater occurrence in the Deccan Traps is controlled by the contrasting water bearing properties of different flow units, thus, resulted in multiple aquifer system, at places. The water bearing zones are the weathered and fractured zones.

Crystalline aquifers

The crystalline hard rock aquifers such as granite, gneisses and high grade metamorphic rocks such as charnockites and khondalites constitute good repository of ground water. Most of the results of groundwater exploration projects have proven that hard rocks neither receive nor transmit water, unless they are weathered and/or fractured. The aquifers are the weathered zone or the fracture system. The fracture system includes fractures, joints,

bedding planes, and solution holes. These openings do not have an even distribution, but are rather localised phenomena. The weathered zone is underlain by semi-weathered rock, fractured rock followed by bedrock. The depth of the bed rock varies from 30-100 m.

In hard rock terrains, ground water occurs under phreatic condition in the mantle of weathered rock, overlying the hard rock while within the fissures, fractures, cracks, joints within the hard rock, ground water is mostly under semi-confined or in the confined state. Compared to the volume of water stored under semi-confined condition within the body of the hard rock, the storage in the overlying phreatic aquifer is often much greater. In such cases, the network of fissures and fractures serves as a permeable conduit feeding this water to the well. Ground water flow rarely occurs across the topographical water divides and each basin or sub-basin can be treated as a separate hydro geological unit for planning the development of ground water resources.

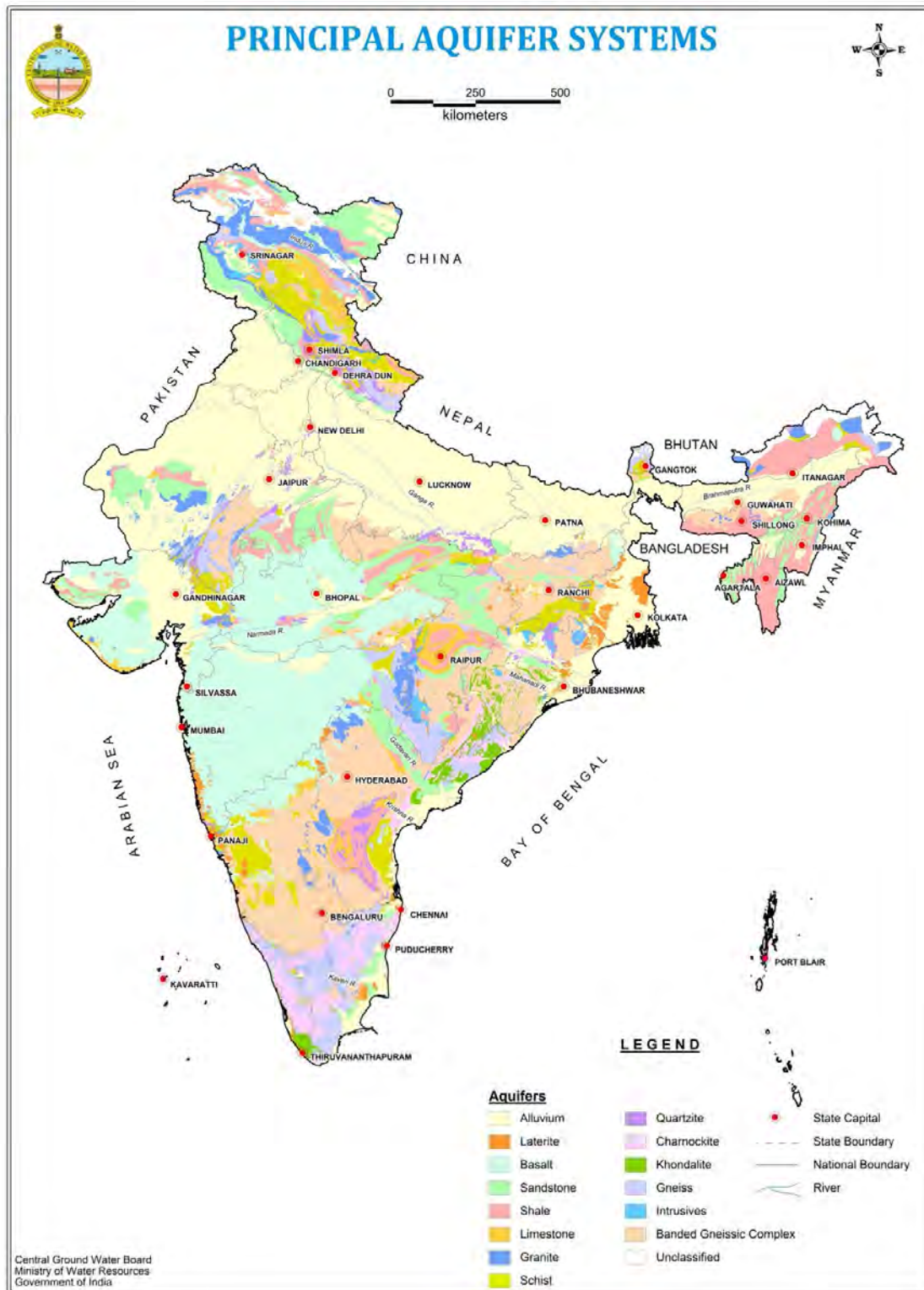


Fig 7 Principal Aquifer Systems of India

4.2 Ground Water Level Scenario in the country

Ground water level is one of the basic data-element which reflects the condition of the ground water regime in an area. Ground water levels are being monitored by Central Ground Water Board and State Ground Water Departments. CGWB monitors ground water level four times a year during January, April/ May, August and November. The periodicity of ground water level monitoring by the State Government varies from State to State. The primary objective of monitoring the ground water level is to record the response of ground regime to the natural and anthropogenic stresses of recharge and discharge parameters with reference to geology, climate, physiography, land use pattern and hydrologic characteristics. The natural conditions affecting the regime involve climatic parameters like rainfall, evapotranspiration etc., whereas anthropogenic influences include pumpage from the aquifer, recharge due to irrigation systems and other practices like waste disposal etc.

CGWB monitors the ground water level and quality in a network of 15640 ground water monitoring wells located all over the country. This data along with State Government monitoring data is used for assessment of ground water resources. The ground water level scenario based on data from CGWB is presented in the following paragraphs.

4.2.1 Ground Water Level Scenario, 2010

The depth to water level map of India for Pre- Monsoon period (May 2010) (**Fig 8**) reveals that that in sub-Himalayan area, north of river Ganges and in the eastern part of the country in the Brahmaputra valley, generally the depth to water level varies from 2-10 meter below ground level (m bgl). In major parts of north-western states (Indus basin), depth to water level generally varies from 10-20 m bgl with pockets of deeper water level of more than 20 m bgl. In the western parts of the country covering the states of Rajasthan and Gujarat deeper water level is recorded in the range of 10-20 m bgl. In western Rajasthan and north Gujarat deeper water level in the

range of 20-40 m bgl and > 40 m bgl have also been also recorded. In the west coast water level is generally less than 10 m and in western parts of Maharashtra State in isolated pockets water level in the range of 2-5 m has also been observed. In the east coast i.e. coastal Andhra Pradesh, Tamil Nadu and Orissa, water level in the range of 2-5 m bgl have been recorded. However South-eastern part of West Bengal recorded water level in the range of 5-10 m bgl. In central India water level generally varies between 5-10 m bgl, with patches where deeper water level more than 10 m bgl has been observed. The peninsular part of country generally recorded a water level in the range 5-10 m bgl. In some patches water level ranges from 10-20 m bgl. Isolated patches of water level of 10-20 m bgl and 20-40 m bgl have been observed. The Post-Monsoon (November 2010) depth to Water level map (**Fig 9**) reveals that in Sub-Himalayan area, north of river Ganges and in the eastern part of the country in the Brahmaputra valley, generally the depth to water level varies from 2-5 meter below ground level (bgl). Isolated pockets of shallow water level less than 2 m bgl have also been observed. In major parts of north-western states depth to water level generally ranges from 10-20 m bgl. In the western parts of the country deeper water level is recorded in the depth range of 20-40 m bgl. In North Gujarat, part of Haryana and western Rajasthan water level more than 40 m bgl is recorded. In the west coast water level is generally less than 5 m and in western parts of Maharashtra State isolated pockets of water level less than 2 m has also been observed. In the east coast i.e. coastal Andhra Pradesh and Orissa, shallow water level of less than 2 m have been recorded. In eastern states, water level in general ranges from 2-5 m bgl. However South-eastern part of West Bengal recorded water level in the range of 5-10 m bgl. In central India water level generally varies between 2-5 m bgl, except in isolated pockets where water level more than 5 m bgl has been observed. Similarly pockets of shallow water level less than 2 m bgl is also observed along the west coast. The peninsular part of country generally recorded water level in the range 2-5 m bgl. In some patches water level ranges from 5-10 m bgl.

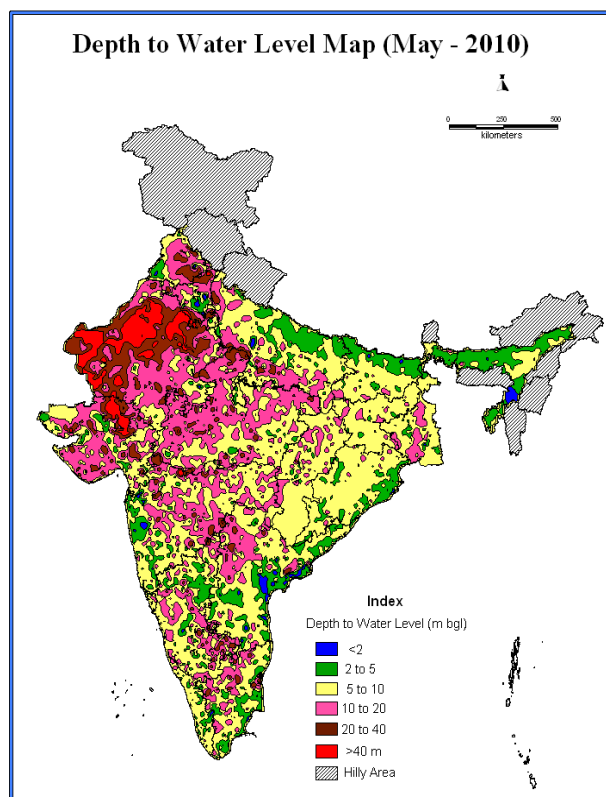


Fig 8 Pre monsoon Depth to Water Level Map-2010

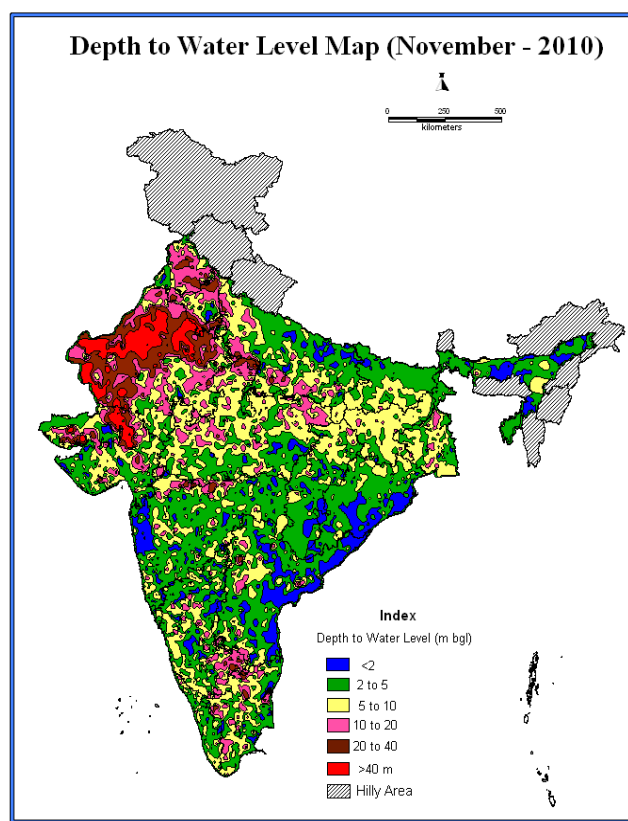


Fig 9 Post- monsoon Depth to Water Level Map-2010

4.2.2 Changes in ground water level regime due to impact of rainfall

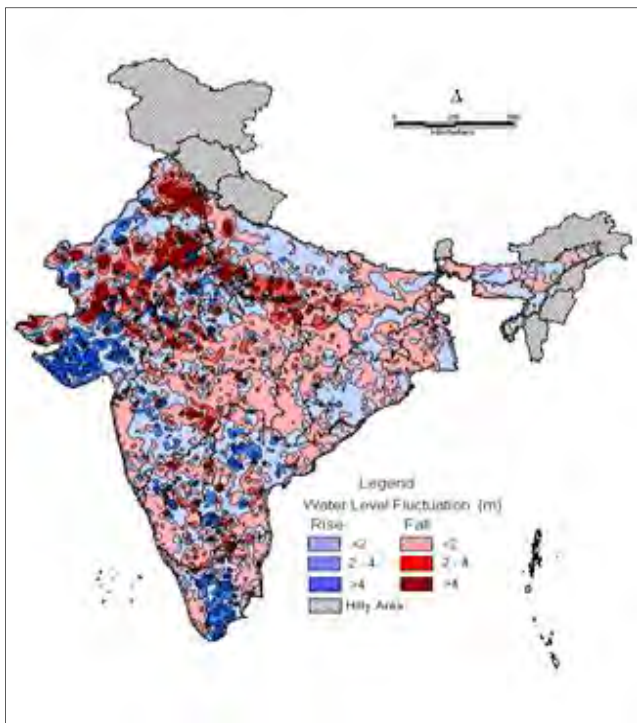
Rainfall is the most significant source of ground water recharge and hence changes in the rainfall pattern leaves distinct imprint on the ground water regime of an area. Since post-monsoon water level indicates the impact of rainfall on ground water recharge, comparison was made between post-monsoon water level of last three years from the year of assessment and the decadal mean water level. Comparison of depth to water level of Post Monsoon 2008 with decadal mean Post Monsoon (1998-2007) (**Fig 10**) reveals that in general, there is a decline in the water levels in south western Uttar Pradesh, Punjab, Haryana, central Rajasthan, south eastern Madhya Pradesh, Chhattisgarh, West Bengal, in parts of Andhra Pradesh, Maharashtra and North Eastern states. Decline in water level of more than 2 m have been observed in south western Uttar Pradesh, Punjab, Haryana and eastern Rajasthan states. Rise in water level is observed in parts of peninsular states, Gujarat and south Rajasthan states. Rise in water level of more than 2 m is prominent in the states of Gujarat, southern and western Rajasthan, Tamil Nadu and in parts in Maharashtra and Andhra Pradesh.

Comparison of depth to water level of Post Monsoon 2009 with decadal mean Post Monsoon (1999-2008)(**Fig 11**) reveals that the decline in water levels has spread to more far flung areas as compared to 2008 in western and southern Uttar Pradesh, Punjab ,Haryana, Rajasthan, Chhattisgarh ,Bihar, Jharkhand, Meghalaya, parts of Bengal etc. This is directly related to rainfall as in 2009, major parts of the country received deficient rainfall except Arunachal Pradesh, Madhya Maharashtra, Coastal Karnataka, Kerala, Tamil Nadu, Puducherry, Konkan, Goa, Gangetic West Bengal which received normal rainfall. The post monsoon rise in water level in 2009 as compared to decadal mean are registered in most parts of Gujarat, Maharashtra, Karnataka, Kerala, parts of Orissa etc.

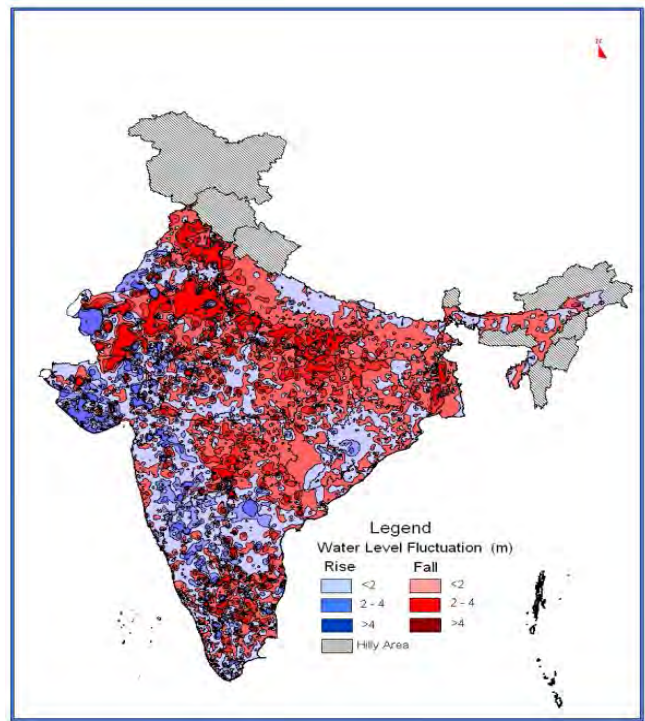
In 2010 most parts of the country received excessive rainfall except Assam, Meghalaya, Bihar, eastern Uttar Pradesh, Jharkhand and Gangetic West

Bengal which received deficient rainfall. This is also reflected in the post monsoon water level fluctuation map of 2010 versus the decadal mean (**Fig 12**).The map shows that most parts of peninsular India, Gujarat, Maharashtra etc have registered a rise in water level whereas the fall in water level in 2010 as compared to decadal mean is conspicuous in Bihar, Jharkhand, West Bengal, Meghalaya, eastern and western parts of Rajasthan etc.

Thus it can be broadly concluded that the changes in ground water level with respect to the mean water level behaves similarly with the changes in rainfall pattern with normal rainfall.



**Fig 10 Water level fluctuation Post-monsoon 2008
Vs
Decadal Mean (1998-2007)**



**Fig 11 Water level fluctuation Post-monsoon 2009
Vs
Decadal Mean (1999-2008)**

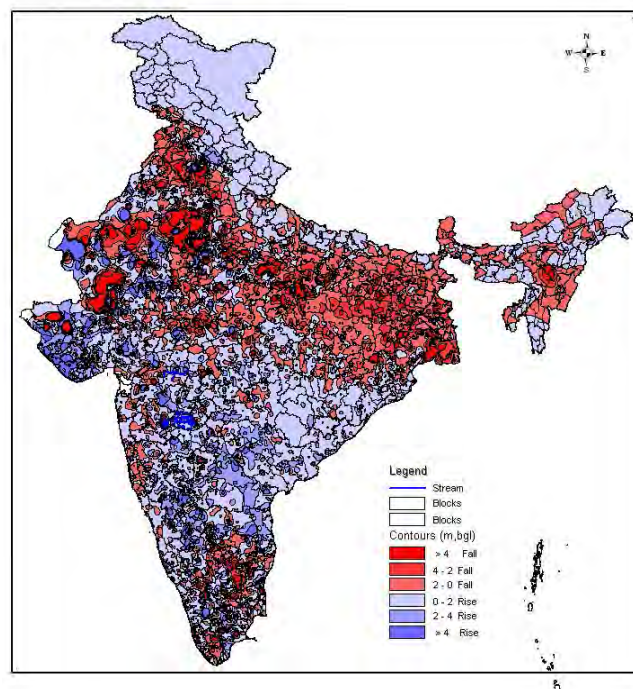


Fig 12 Water level fluctuation Post-monsoon 2010 vs Decadal Mean (2000.-2009)

CHAPTER 5

GROUND WATER RESOURCES OF INDIA

5.1 Dynamic Fresh Ground Water Resources

The dynamic ground water resources of the States and Union Territories have been assessed jointly by the CGWB and State Ground Water Departments under the supervision of the State level Committees. The base year of computation of the resources is 2010-11. The ground water resources in some of the State/Union Territory/Islands viz. Sikkim, Andaman & Nicobar, Dadara & Nagar Haveli, and Lakshadweep have been estimated by CGWB in absence of active participation of State Govt. The ground water assessment figures computed at the State Level are presented in the following compilation.

The dynamic ground water resources are also known as Annual Replenishable Ground Water Resources since it gets replenished/ recharged every year. The Annual Replenishable Ground Water Resource for the entire country has been assessed as 433 billion cubic meter (bcm). The major source of ground water recharge is the monsoon rainfall. About 58% of the annual replenishable resources i.e. 253 bcm are contributed by monsoon rainfall recharge. The overall contribution of rainfall to country's Annual Replenishable Ground Water Resource is 68% and the share of other sources viz. canal seepage, return flow from irrigation, recharge from tanks, ponds, and water conservations structures taken together is 32%. State-wise Ground Water Resources of India as on March, 2011 is given in **Annexure – I** and the district-wise figures are given in **Annexure - II**. **Figure-13** presents the over-all scenario of ground water resource utilization and availability of the country. The contribution from other sources such as canal seepage , return flow from irrigation, seepage from water bodies etc in Annual Replenishable Ground Water Resource is more than of 33% in the states of Andhra Pradesh, Delhi, Haryana, Gujarat, Goa, Jammu & Kashmir, Karnataka, Punjab,

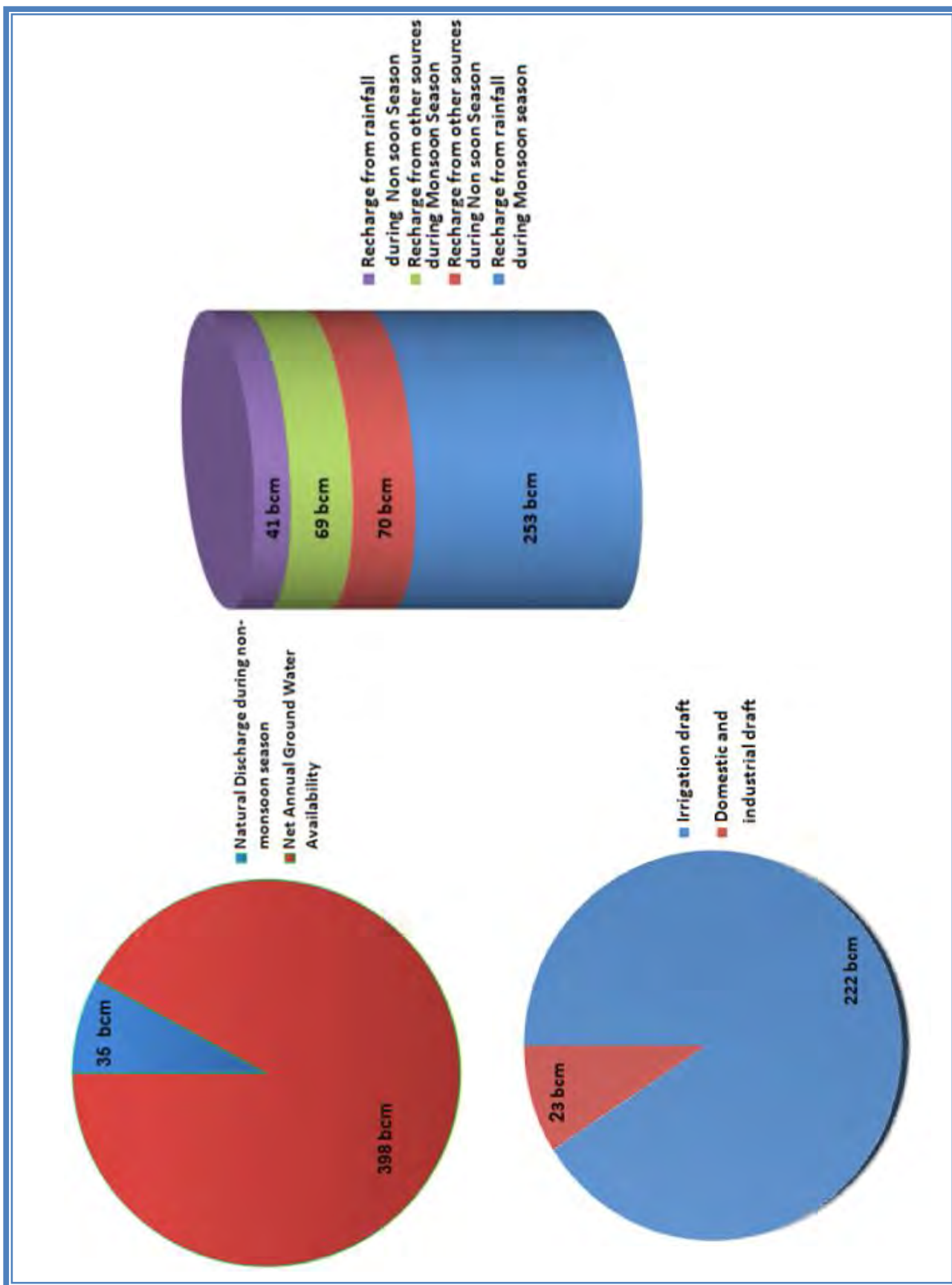
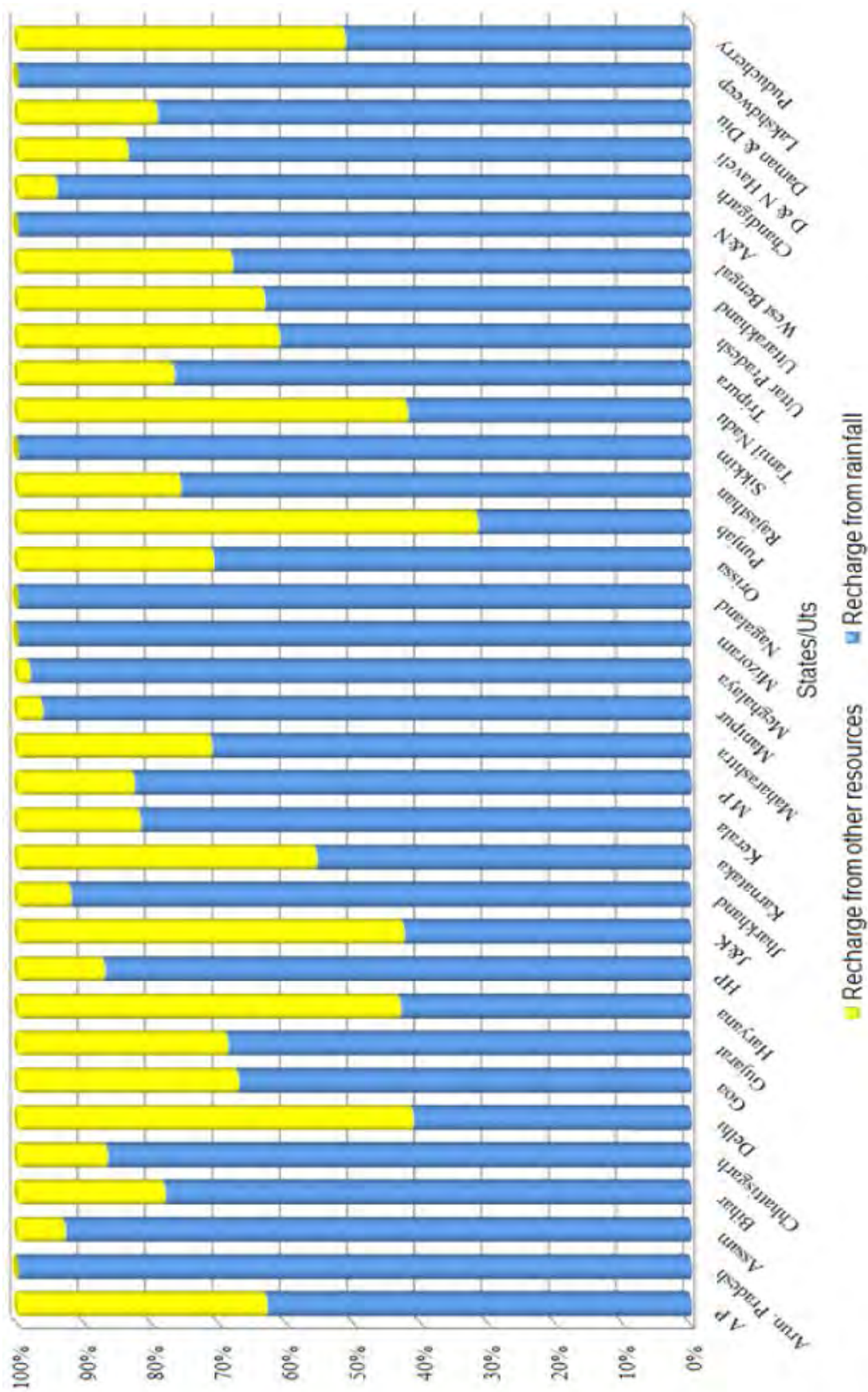


Fig.13 Ground water Resources Availability and Utilization in India

Fig.14

Recharge from Rainfall vis-a-vis Other Sources

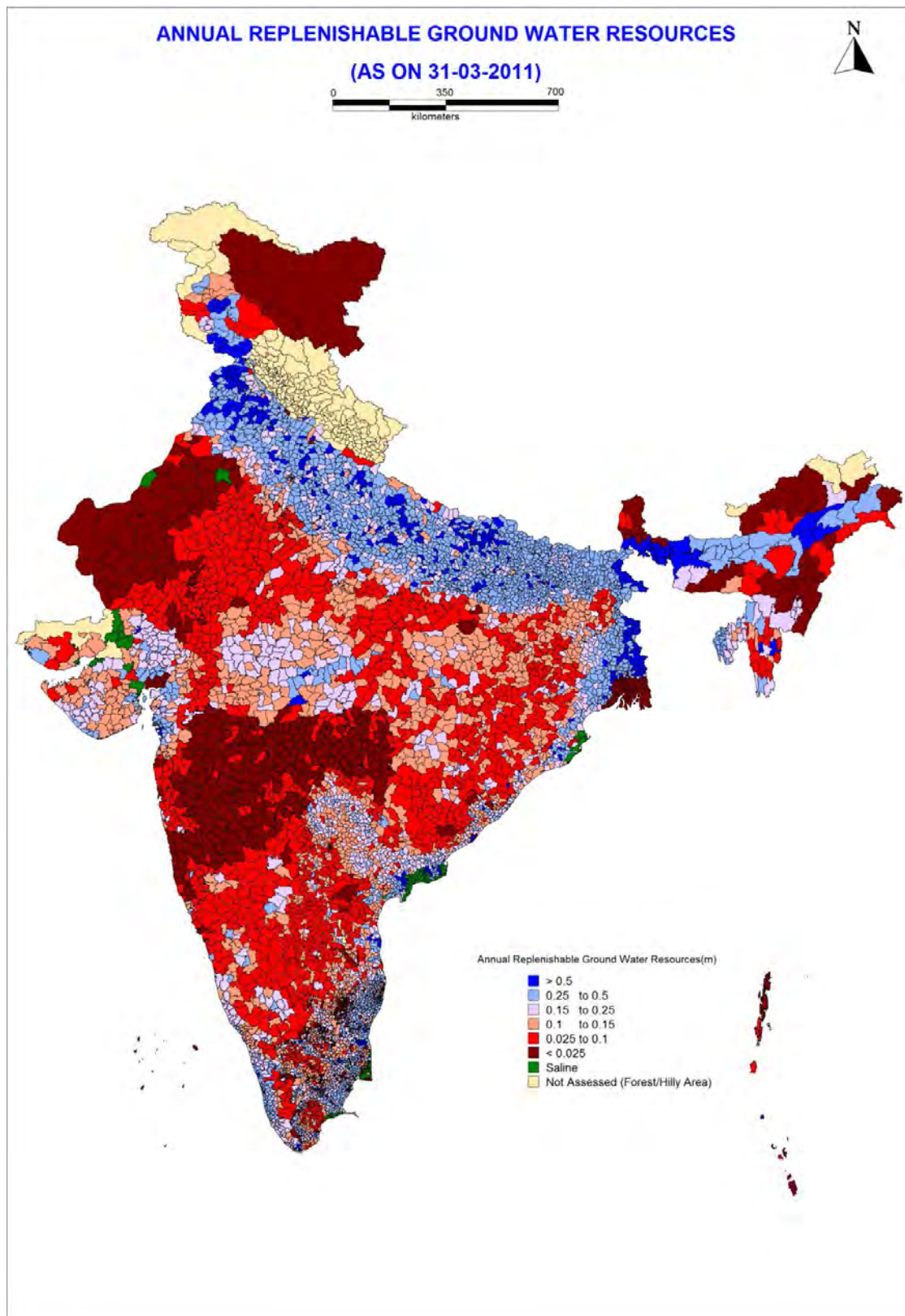


Tamil Nadu, Uttar Pradesh, and UT of Puducherry (**Figure-14**). South-west monsoon being the most prevalent contributor of rainfall in the country, about 73% of country's Annual Ground Water Recharge takes place during the Kharif period of cultivation. Keeping 35 bcm for natural discharge, the Net Annual Ground Water Availability for the entire country is 398 bcm.

The spatial variation in annual replenishable ground water resources is presented in **Figure-15**. Volumetric estimates are dependent on the areal extent of the assessment unit. Thus, relative comparison of ground water resource of different assessment units based on volumetric estimates is not possible. Hence volumetric estimates of annual replenishable ground water resources have been divided by the area of the assessment unit to arrive at estimates per unit area (in meter). Replenishable Groundwater resource is significantly high in the Indus–Ganga–Brahmaputra alluvial belt in the North, East and North East India covering the states of Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal and valley areas of North Eastern States, where rainfall is plenty and thick piles of unconsolidated alluvial formations are conducive for recharge. Annual Replenishable Ground Water Resource in these regions varies from 0.25 to more than 0.5 m. The coastal alluvial belt particularly Eastern Coast also has relatively high replenishable ground water resources, in the range 0.25 to more than 0.5 m. In western India, particularly Rajasthan and parts of northern Gujarat which have arid climate, the annual replenishable ground water resources are scanty, mostly up to 0.025 m. Similarly, in major parts of the southern peninsular India covered with hard rock terrains, annual replenishable ground water recharge is less, only up to 0.10 m. This is primarily because of comparatively low infiltration and storage capacity of the rock formations prevailing in the region. The remaining part of Central India is mostly characterized by moderate recharge in the range of 0.10–0.25 m.

The overall estimate of annual replenishable ground water resources of the entire country shows a marginal increase in the present estimate as

Figure-15



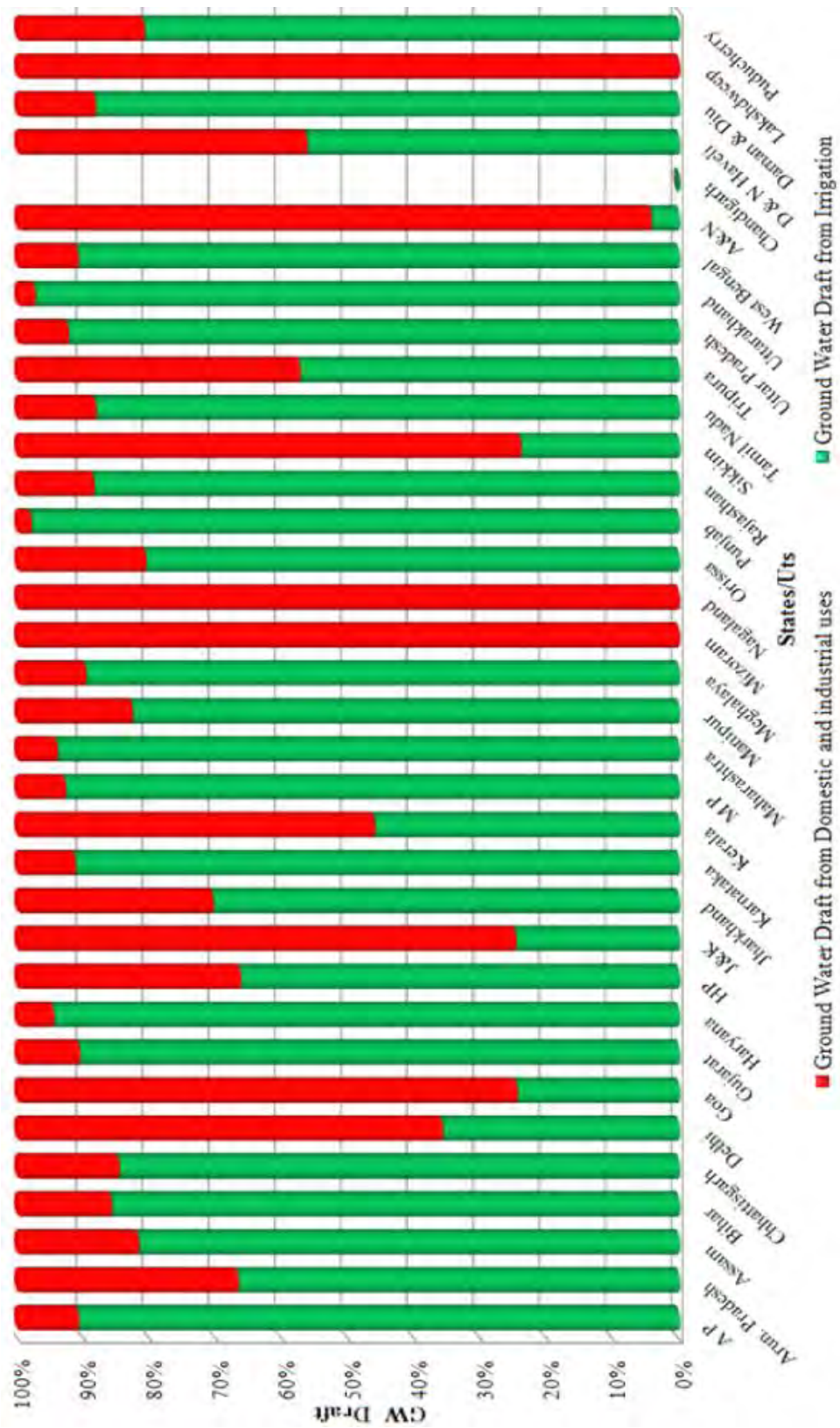
compared to the 2009 by about 2 bcm. However there are significant variations in the recharge estimates of some of the States as indicated in Section 5.5. The main reasons for this can be attributed to – changing ground water regime, widespread implementation of rainwater harvesting and water conservation measures, changes in rainfall pattern, adoption of revised values of parameters like Specific Yield which were estimated based on subsequent field studies and availability of improved database which helped in refinements in assessment.

5.2 Ground Water Utilization

The assessment of ground water draft is carried out based on the Minor Irrigation Census data and sample surveys carried out by the State Ground Water Departments. The Annual Ground Water Draft of the entire country for 2010-11 has been estimated as 245 bcm. Agriculture sector remained the predominant consumer of ground water resources. About 91% of total annual ground water draft i.e. 222 bcm is for irrigation use. Only 23 bcm is for Domestic & Industrial use which is about 9% of the total draft. An analysis of ground water draft figures indicates that in the states of Arunachal Pradesh, Delhi, Goa, Himachal Pradesh, Jammu & Kashmir, Jharkhand, Kerala and Odisha, Sikkim, North Eastern states of Manipur, Mizoram, Nagaland and Tripura, Sikkim, and Union Territories of Andaman & Nicobar Island, Dadra & Nagar Haveli, Lakshadweep and Puducherry, ground water draft for domestic & industrial purposes are more than 15% (**Figure-16**).

There has been marginal (2 bcm) increase in the overall estimate of the annual ground water draft of the country in 2011 as compared to 2009. However, in some States, decrease in ground water draft estimates has also been recorded which have been attributed to lower yield of bore wells in Over-exploited blocks in hard rock terrain, good rainfall and provision for alternate source of water resulting in lesser stress on ground water resources etc. State-wise details are given in Section 5.5.

Figure.16 Irrigation draft vis-à-vis domestic industrial draft



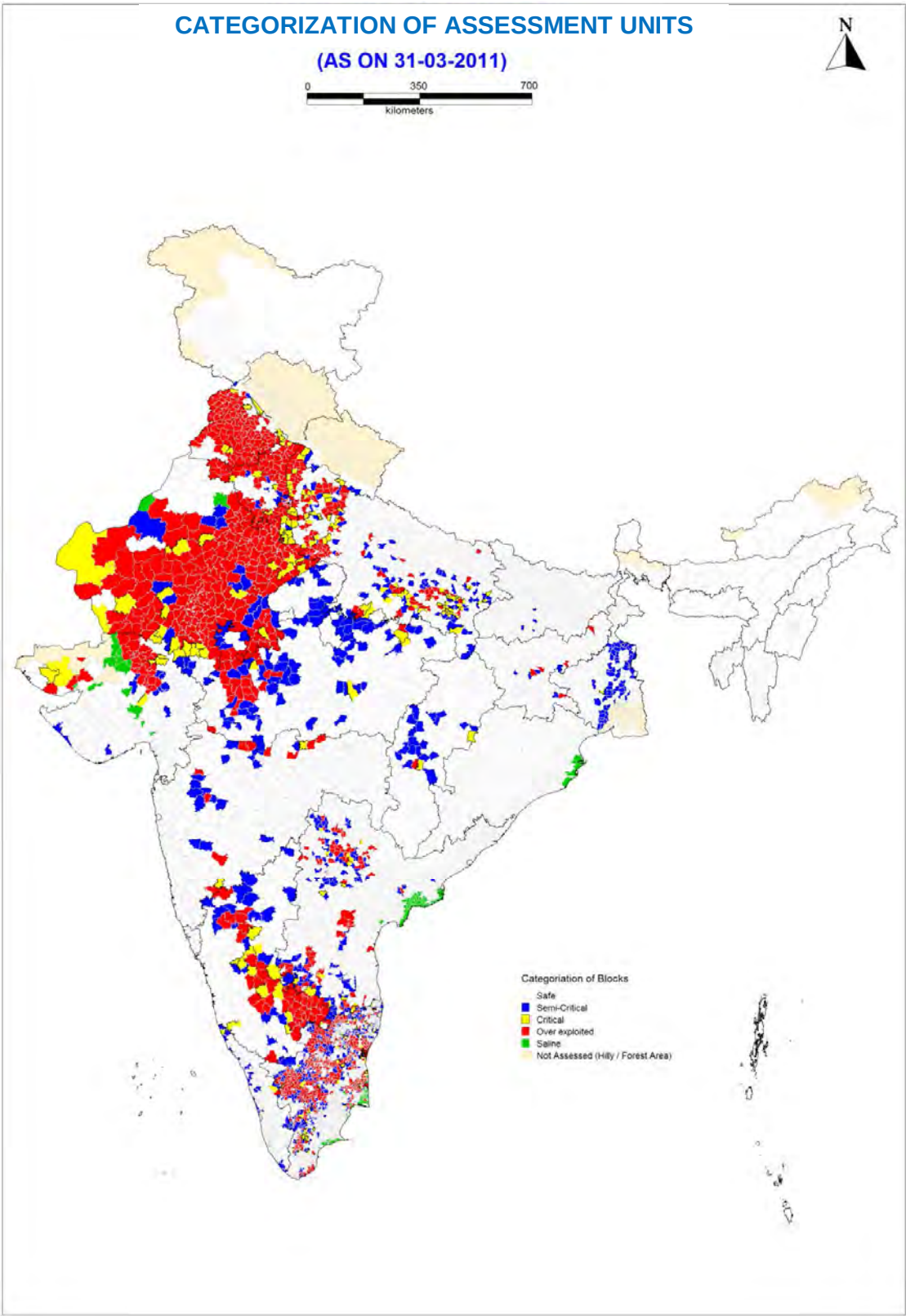
5.3 Stage of Ground Water Development

The overall stage of ground water development in the country is 62%. The status of ground water development is very high in the states of Delhi, Haryana, Punjab and Rajasthan, where the Stage of Ground Water Development is more than 100%, which implies that in the states the annual ground water consumption is more than annual ground water recharge. In the states of Himachal Pradesh, Tamil Nadu and Uttar Pradesh and UTs of Daman & Diu, and Puducherry, the stage of ground water development is 70% and above. In rest of the states / UTs the stage of ground water development is below 70%. The ground water development activities have increased generally in the areas where future scope for ground water development existed. This has resulted in increase in stage of ground water development from 61% (2009) to 62% (2011).

5.4 Categorization of Assessment Units

Out of 6607 numbers of assessed administrative units (Blocks/ Taluks/ Mandals/ Districts), 1071 units are Over-exploited, 217 units are Critical, 697 units are Semi-critical, and 4530 units are Safe. Apart from these, there are 92 assessment units which are completely Saline (Annexure – III). Number of Over-exploited and Critical administrative units are significantly higher (more than 15% of the total assessed units) in Delhi, Haryana, Himachal Pradesh, Karnataka, Punjab, Rajasthan and Tamil Nadu, Uttar Pradesh and also the UTs of Daman & Diu and Puducherry (**Figure-17**).

Figure-17



A perusal of the Categorization map indicates that the Over-exploited blocks are concentrated in the North Western, Western and Southern Peninsular part of the country. The reason for over-exploitation in the North Western part i.e. Punjab and Haryana is indiscriminate extraction of ground water mainly for irrigation purpose. In the Western part of the country viz. Rajasthan and Gujarat, over-exploitation is caused by arid climate resulting scanty and irregular rainfall and consequent less recharge. In the southern part of the country i.e. Karnataka, and Tamil Nadu, large number of over-exploited blocks are caused because of hard rock terrain which permits less recharge and thus result in water stressed conditions.

List of categorization of Blocks / Taluks/ Mandals/ Districts is given in **Annexure – IV**.

5.5 State-wise Ground Water Resources Scenario

The ground water conditions, its availability and utilization scenario in various States are described in the following paragraphs.

ANDHRA PRADESH

Andhra Pradesh state covers an area of 2, 75,068 km² with coastal line of 974 km and forest cover of 23%. The state shares its boundaries with Orissa, Tamilnadu, Chattisgarh, Maharashtra and Karnataka and on the eastern side bounded by the Bay of Bengal. The state is divided in to 1229 watersheds as the State is predominantly covered by hard rocks. The ground water resources of these watersheds were estimated separately for Command, Non-command and Poor Quality areas. The total area considered for recharge is 2, 38,001 km² and out of this 53,744 km² falls in command area and 1, 76,700 km² in non-command area and remaining 7,557 km² in poor ground water quality area. The state is underlain by diverse rock types of different geological ages from Pre-Cambrian to Recent. As much as 85% of the State is underlain by consolidated formations like Archaeans, Cuddapahs, Dharwars, Kurnools, Deccan Traps etc. Rest of the area is underlain by soft rocks including Gondwanas, Rajahmundry sandstone and

Alluvium. Ground water in phreatic zone is generally through open wells, filter points and cavity wells.

The Ground water resources have been assessed watershed wise and were apportioned to mandals. The Annual Replenishable Ground Water Resource of the State has been estimated as 35.89 bcm and Net Annual Ground Water Availability is 32.57 bcm. The Gross Ground Water Draft for all uses is estimated as 14.51 bcm and Stage of Ground Water Development is 45%. Out of 1110 blocks, 83 blocks have been categorized as 'Over-exploited', 15 as 'Critical', 97 as 'Semi-Critical', 877 as 'Safe' and 38 as saline blocks in the state. Comparatively high development of ground water has been observed in Medak, Anantapur, Chittoor, Rangareddy, Nizamabad, Kaimnagar, Warangal and Prakasam districts. As compared to 2009 estimates, the Net Ground Water Availability and Gross Ground Water Draft as in 2011 were increased from 30.76 bcm to 32.57 bcm (5.88% rise) and 14.15 bcm to 14.51 bcm (2.5% rise) respectively, resulting in the reduction of Stage of Ground Water Development from 46% in 2009 to 45% in 2011. The changes in the recharge and categorization in respect of assessment units are mainly due to good monsoon rainfall in the year 2010, updation of data on water harvesting structures and usage of low unit draft values due to availability of more rain water for the agriculture.

ARUNACHAL PRADESH

The state is underlain by diverse rock types of different geological ages from Pre-Cambrian to Recent and more than 90% of the area is covered by hilly terrain. Major part is covered with consolidated crystalline rocks and meta-sediments of Precambrian and Palaeozoic times, while Tertiary sediments consisting semi-consolidated argillaceous assemblage, represented by the Disang, Barail, Tipam, Siwalik and Dihing groups of rock, occupy periphery areas bordering Assam and behave as run-off, infiltration as also discharge zones. In consolidated formations ground water potentiality appears to be very much limited. Semi-consolidated Tertiary formations are likely to give moderate or poor yield and expected to be controlled by aquifer geometry

and structural features. Ground water in both consolidated and semi-consolidated formations is manifested as springs. Springs in all geological formations are both seasonal and perennial in nature.

Unconsolidated Quaternary sediments comprising the terrace deposits of Pleistocene (Bhabar zone) and also the terrace and alluvial fan deposits of Holocene age prevail in the fringe valley areas and as thin carpet in isolated structural valleys and with considerable thickness in open and wide valleys joining Brahmaputra Alluvial plains. The unconsolidated alluvial sediments in the valley areas act as good repositories for ground water development. Valleys adjoining Assam are most promising where good thickness of granular zones is distributed. Discharge of the deep tube wells, tapping mostly unconsolidated Quaternary sediments & at places Upper Tertiary formations, varies from 1.4 m³/hr to 54 m³/hr, while transmissivity ranges from 1.14 to 661 m²/day. Storativity ranges from 0.35 x 10⁻³ to 6.65 x10⁻³. Autoflow conditions seen at places are promoted due to high hydraulic head.

Ground water resources have been assessed district-wise and the resources of eleven districts, out of the total 16 district of the State, have been assessed for the assessment year 2010—11 and the ground water resource potential of the remaining five districts, namely, Upper Siang, Anjaw, Dibang Valley, Kurung Kumey and Tawang could not be estimated due to paucity of data. The Annual Replenishable Ground Water Resource of the State has been estimated as 4.51 bcm and Net Annual Ground Water Availability is 4.06 bcm. The Annual Ground Water Draft is 0.003 bcm and Stage of Ground Water Development is 0.08%. All the districts have been categorized as 'Safe' and there is no saline area in the state. As compared to 2009 estimate, there is no significant change in annual ground water recharge and ground water draft.

ASSAM

The State is underlain mainly by unconsolidated Quaternary formation in Brahmaputra valley and potential aquifers lie at shallow as well as deeper zone. The yield potential of deep tube wells in the area ranges from 100 to 300 m³/hr for negligible drawdown within 8 m and transmissivity ranges from

2000 m²/day to more than 4000 m²/day and permeability range varies from 20 m/day to 60 m/day. The semi-consolidated Tertiary formations are found to occur in the southern part of Karbi Anglong, Cachar, Karimganj and Hailakandi districts and in Upper Assam covering southern fringe of Dibrugarh, Tinsukia, Sibsagar, Jorhat, Golaghat districts. The tube wells are feasible with yield potential within 50 to 100 m³/hour for drawdown within 20 m. The consolidated Precambrian rocks occurs mainly in N.C. Hills, Karbi-Anglong, Kamrup, Goalpara, Dhubri, Nagaon districts and the bore wells may yield <15 lps for considerable drawdown.

Ground water resources have been assessed district-wise due to paucity of block wise data. The Annual Replenishable Ground Water Resource of the State has been estimated as 28.52 bcm and Net Annual Ground Water Availability is 25.79 bcm. The Annual Ground Water Draft is 3.49 bcm and Stage of Ground Water Development is 14%. All the districts have been categorized as 'Safe' and there is no saline area in the state. As compared to 2009 estimate, the Net Ground Water Availability and Gross Ground Water Draft as in 2011 have decreased by 2.02 bcm & 2.54 bcm respectively, resulting in the Stage of Development decreasing by 8% during 2011 than that of 2009. The changes in the recharge and draft parameters in respect of assessment units can be attributed to decrease in recharge components from ground water irrigation and the ground water draft for irrigation use during 2011, which have been calculated considering the fourth census data of 2006-07 and projected 2% per year. As per the fourth census, the number of shallow tube well became half of the figure during 2004 and gross ground water draft for irrigation has markedly decreased by 2.47 BCM in 2011 in comparison to 2009 which was calculated considering the projected figure, based on the figure of 2004.

BIHAR

The state is covered with Gangetic alluvium in more than 89% of its geographical area. The consolidated formations occupy fringes in the southern parts of the state. Dug wells and shallow tube wells tapping the

phreatic zone are the common groundwater abstraction structures. The yield of the wells generally ranges from 3-10 lps.

With 'Block' as the assessment unit, the estimation of dynamic ground water resources has been carried out for 533 blocks in the state. The Annual Replenishable Resource has been worked out as 29.34 bcm with the Net Groundwater Availability as 26.86 bcm. The Gross Ground Water Draft for all uses has been estimated as 11.95 bcm. In comparison to the previous estimation of 2009, the Net Ground Water Availability and Gross Ground Water Draft as on 2011 have increased by 2.52% and 5.19% respectively. The Stage of Ground Water Development of the state is 44.5%, which was earlier at 43% in the 2009 estimate. A perusal of the Stage of Ground Water Development indicates that out of 533 blocks in the state, 522 blocks fall under 'Safe' category where available water level trend of wells do not reveal any significant lowering for any patch with significant aerial extent. The rest 11 blocks has been categorized as 'Semi-Critical'. Out of these, 6 blocks fall in the South Bihar Plains and 3 blocks fall in the North Bihar plains. No block has been categorised as critical or over-exploited.

CHHATTISGARH

The state is underlain by diverse rock types of different geological ages from Pre-Cambrian to Recent. 87% area of the State is underlain by hard rocks being tapped mostly by dug wells constructed in the weathered zone and bore wells to tap the deeper aquifers. The yield of open (dug) wells varies from 1 to 2 lps and the yield of the bore wells ranges from < 1 to 5 lps. 13% area of the State is occupied by Semi-consolidated sedimentary rocks where Dug wells & tube wells have yield range of 1 to 10 lps.

The Ground water resources have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 12.42 bcm and Net Annual Ground Water Availability is 11.63 bcm. The Annual Ground Water Draft is 4.05 bcm and Stage of Ground Water Development is 35%. Out of 146 blocks, 18 have been categorized as Semi-

critical, 2 as Critical, one as overexploited and remaining 125 as Safe. In Chhattisgarh, the ground water development concentrates in the central part of the state i.e. within Chhattisgarh basin more as compared to the other parts of the state. Therefore, most of the Semi-critical, critical and over exploited blocks are falling in the Central part of the State. As compared to 2009 estimate, there is increase in Annual Replenishable Ground Water Resources in 2011 by about 1.63%, which is mainly due to the increase in recharge from other sources. The Ground Water Draft has been increased by about 11.27% during current assessment as compared to draft during 2009.

DELHI

The state is covered by diverse rock types of different geological ages from Pre-Cambrian to Recent. As much as 89% of the State is occupied by alluvium and ground water is being tapped mostly through tube wells. Yields of tube wells vary from 4 to 10 lps in older alluvial deposits and from 25 to 55 lps in newer alluvium. About 11% of the State is occupied by quartzitic hard rock where bore wells have yields of 0.6 to 5 lps.

The Ground water resources have been assessed tehsil-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 0.31 bcm and Net Annual Ground Water Availability is 0.29 bcm. The Annual Ground Water Draft is 0.39 bcm and Stage of Ground Water Development is 137%. Out of 27 tehsil, 18 tehsils have been categorized as 'Over-exploited', two as 'Critical', 5 as 'Semi-Critical', 2 as 'Safe'. Comparatively high development of ground water has been observed in all districts except Central, New Delhi and districts. As compared to 2009 estimate, the Gross Ground Water Draft as in 2011 have decreased by about 1%, resulting in the Stage of Ground Water Development at about 137%. The changes in the recharge and draft parameters in respect of assessment units are due mainly to the reduction in domestic Ground Water Draft, during the current assessment is because of the actual population as per 2011 Census, whereas in 2009 the projected population based on 2001 Census figure was taken for estimation of draft. The slightly increase in surface water supply by DJB, the

regulation on drilling of new bore wells in whole NCT Delhi also contributed to lesser dependency on Ground Water.

GOA

Major part of the Goa State is covered by consolidated formation of Dharwar Super Group. Ground water occurs under unconfined to semi-confined conditions in beach sands, laterites and weathered and fractured crystalline rocks. The development of ground water from phreatic zone is mostly through dug wells and shallow bore wells.

Ground Water Resource has been assessed taluk-wise. The Annual Replenishable Ground Water Resource has been estimated as 0.24 bcm and Net Annual Ground Water Availability is 0.145 bcm. The Annual Ground Water Draft is 0.041 bcm and Stage of Ground Water Development is 28% and all the taluks in the state have been categorized as Safe.

There is a marginal increase in the estimates of annual replenishable resources and net annual ground water availability as compared to the previous 2009 assessment. The main reason for this is 43 % reduction in number of dugwells even though there is a marginal increase of 10% in number of bore wells.

GUJARAT

The state is underlain by diverse rock types of different geological ages from Pre-Cambrian to Recent. As much as 60 % of the State is underlain by hard rocks and rest by soft rock / alluvium formations. In hard rock areas, the ground water is tapped mostly through dug wells constructed in the weathered zone. Dug cum bore wells and deep bore wells are common for irrigation. In alluvium & soft rock areas, deep tube wells are common for both irrigation and domestic usage. A yield of open (dug) wells varies from 2 to 10 m³/day whereas, that of tube wells ranges from less than 10 to 100 m³/day. Most of the alluvium areas of central and north Gujarat have experienced high groundwater development since last two decades. Majority of the overexploited block are in the alluvial aquifer areas. The State has 1600 km long coast line and many areas along it are facing salinity ingress problem.

The Ground water resources have been assessed on taluka –wise basis. The Annual Replenishable Ground Water Resource of the State has been estimated as 18.57 bcm and Net Annual Ground Water Availability is 17.59 bcm. The Annual Ground Water Draft is 11.86 bcm and Stage of Ground Water Development is 67 %. Out of 223 assessment units (taluka area), 24 talukas have been categorized as ‘Over-exploited’, 5 as ‘Critical’, 13 as ‘Semi-Critical’, 171 as ‘Safe’ and there are 10 saline talukas in the state. Comparatively high development of ground water has been observed in Banaskantha, Mahesana, Patan, Gandhinagar, Ahmedabad and Kachchh districts. As compared to 2009 estimate, the Net Ground Water Availability have increased by about 1.36 % whereas Gross Ground Water Draft have decreased by 8.73 % in 2011, resulting in lowering of Stage of Ground Water Development from 74.87 % in 2009 to 67 % in 2011. The changes in the recharge and draft parameters in respect of assessment units are due to the increased in recharge activities, more rainfall in last five years and also due to adoption of water conservation methods like drip and sprinkler irrigation in many of over exploited blocks which has decreased the irrigation draft leading to modification of parameter values.

HARYANA

Haryana State is occupied by the alluvial deposits, which covers around 98% of the state while hard rocks cover just around 2%. Alluvial deposits are of Older and Newer types and consist chiefly of clay, silt and fine to medium sand. Other deposits are piedmont deposits, which are confined to a narrow zone, about 2 to 4 kms wide, between Siwalik hills and alluvial plains. Sand-dunes are found in the districts of Bhiwani, Mahendragarh, Hissar and Sirsa. Coarse sand, gravels and boulders are found to occur in piedmont areas and in the adjacent alluvial tracts. The hard rock formations belong to the formation of Delhi systems of Pre-Cambrian age and occupying the southern part of the state, while Shivalik system of Tertiary age are occupying the northern most part of the state. Broadly, three-aquifer groups down to the depth of 450 m. bgl. have been deciphered by Central Ground Water Board

in the depth range of 40 to 167 mbgl., 65 to 294 m bgl. and 197 to 383 mbgl. in Ambala-Karnal-Panipat region. In June 2010, 40% of the area the depth to water in general lies within 10 mtrs below ground level which was 53% during June, 2008.

The Ground water resources have been assessed block-wise for the period 2010-11 (as on March 2011). The Annual Replenishable Ground Water Resource of the State has been estimated as 10.78 bcm and Net Annual Ground Water Availability is 9.79 bcm. The Annual Ground Water Draft is 13.05 bcm and Stage of Ground Water Development is 133%. Out of total 116 assessed blocks taken for study, 71 Blocks (61%) are over-exploited, 15 Blocks (13%) are critical, 07 Blocks (6%) are semi critical and 23 Blocks (20%) are in safe category. The analysis of present ground water resource assessment indicates that there is marginal decrease in net availability of ground water resources as compared to previous assessment carried out for the period 2004-08 (as on March 2009). The ground water draft has increased by about 5 %. It has been observed that 9 blocks have shown change to higher category owing to increased ground water draft for irrigation and other uses. 15 blocks have shown change to lower category due to increased replenishable recharge, reduction in draft of tubewells, rise in water level trends or change in the land use.

HIMACHAL PRADESH

The diverse physiographic, climatic, topographic and geologic conditions have given rise to diversified ground water situation in different parts of the state. The rock formation ranges in age from Archean to Recent occupy the area and control the occurrence and movement of ground water depending upon aquifer composition, structure and deposition. Hilly and mountainous parts with steep slopes mainly constitute the run off areas and have low ground water potential. In valley and low-lying areas, unconsolidated / semi-consolidated formations form good potential aquifers.

In consolidated formations the water availability is restricted to weathered mantle, joints/fractures, weak planes, bedding planes and

limestone caverns. The limestones are associated with phyllites and quartzite forms potential aquifers. In granites, ground water availability is minimum in the weathered zone of 5-10m and potentiality of the aquifer is highly dependable on the fracture intensity. The discharges of these granitic aquifers are in the range of 1-3 lps. Groundwater in hard rock areas is either developed through bore wells or springs. Springs are yielding sometimes more than 40 lps and utilized for both drinking and irrigation purposes. Weathered mantle in low topography areas form poor aquifers. Borewells are also constructed in oozing out spring zones for collecting the water to fulfill the domestic water needs.

In the unconsolidated formations the occurrence and movement of ground water is highly dependent on lithology particularly the percentage of clay. These are confined to valley fill areas, regionally extensive, having prolific yield prospects that can sustain moderate to high capacity deep tube wells. The yield potentials of the tube wells depends on the thickness of the total granular zones available within the aquifers tapped and ranging from 5-40 lps in different valleys.

The Ground water resources have been assessed valley-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 0.56 bcm and Net Annual Ground Water Availability is 0.53 bcm. The Annual Ground Water Draft is 0.38 bcm and Stage of Ground Water Development is 71%. Out of 8 valleys, one valley has been categorized as 'Over-exploited', two as 'Critical', five as 'Safe' and there is no saline valley in the state. Comparatively high development of ground water has been observed in Kala Amb valley, Sirmaur district, Una valley & Hum valley of Una district. As compared to 2009 estimate, the Net Ground Water Availability and Gross Ground Water Draft as on March, 2011 have increased by 15% and 22.6% respectively, resulting in increase in the Stage of Ground Water Development by 6.5%. The changes are mainly attributed to refinements in the recharge and draft parameters.

JAMMU & KASHMIR

The dynamic ground water resources of Jammu & Kashmir have been estimated for valley areas and outer plains of all districts of the State. The total recharge of ground water involves several components and the rainfall being the major one. The other components are seepage from canal and return irrigation flow from surface water and ground water.

The Annual Replenishable Ground Water resource of the State has been estimated as 4.25 bcm and Net Annual Ground Water Availability is 3.83 bcm. The Annual Ground Water Draft is 0.81 bcm and the Stage of Ground Water Development is 21%. The entire assessment unit falls under Safe Category. As compared to the 2009 estimates, the Net Ground Water Availability has increased by 15% and the Gross GW Draft has increased by 10%. The change in the Net Ground Water Availability is due to the refinements in the parameter.

The high ground water development in Srinagar can be attributed to low recharge due to the presence of Karewas, which has low specific yield, less rechargeable valley area and small valley portion of Udhampur district.

Assessment units are categorized for Jammu and Kathua districts based on both stage of ground water development and long term trend of water levels. For rest of the districts, state of ground water development is the sole criteria for categorization.

JHARKHAND

The state is underlain by diverse rock types of different geological ages ranging from, Archaean to Recent. The major rock type is Igneous & Metamorphic Rocks covering nearly 85 percent of the geographical area of the state. The weathered zone ranges generally between 10-25 m, is a good repository of ground water; however the recent exploratory wells drilled by CGWB revealed that the secondary porosities below the weathered zones also form potential aquifers. The yield of the exploratory wells range from negligible up to 151 m³/hr. The Vindhyan rocks do not form potential aquifer system as such. The yield of the dug wells (depth range 5.5 to 11m) recorded

from 0.5 to 0.75 m³/hr. The dug wells in volcanic rocks tapping, the weathered mantle have an average yield of 0.5 to 1.2 m³/hr. In Gondwana Supergroup discharge of borewells is 7-10 m³/hr, was obtained from 2 to 3 fractures within 200m bgl. In Tertiary deposits yield ranges from 18 to 78 m³/hr. The Younger Alluvium deposits are confined in patches. The depth of dug wells in general ranges between 10 – 15 m bgl while the depth of shallow tube wells varies between 20 – 40 m bgl and discharge may go up to 40 m³/hr if sediments are available with sufficient thickness.

The Ground water resources have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 6.31 bcm and Net Annual Ground Water Availability is 5.76 bcm. The Annual Ground Water Draft is 1.86 bcm and Stage of Ground Water Development is 32%. Out of 210 blocks, six blocks has been categorized as 'Over-exploited', five as 'Semi-Critical', 199 as 'Safe' and there are no saline block in the state. Comparatively high development of ground water has been observed in Dhanbad, Godda, Ranchi and Hazaribagh districts. As compared to 2009 estimate, the Net Ground Water Availability and Gross Ground Water Draft as in 2011 have increased by about 6.5% and 15.53% respectively, resulting in the increase in Stage of Ground Water Development from 30% to 32%. The increase in the recharge and draft parameters in respect of assessment units are due mainly to the increase Domestic & Industrial Water Supply and in the irrigation draft.

KARNATAKA

Karnataka state is located in southwestern part of India and covers an area of 191761 sq km. It is divided into 30 districts and 176 taluks. The state is drained by five river basins namely Godavari, Krishna, Cauvery, Pennar and West Flow rivers. Though the state is blessed with the bounties of nature, 63% of the land falls under dry tracts ranking second only to Rajasthan in having arid tracts. Major part (99%) of the Karnataka is occupied by hard rock aquifers leaving a small part of the coastal terrain, which is occupied by alluvium. The aquifer systems are classified into nine major groups depending

upon their characteristics and distribution namely Banded Gneissic Complex (BGC), Basalt, Schists, Granites, Charnockites, Limestones, laterites, Sandstones and alluvium. The state is dependent on ground water for its all round development. There is a spurt in groundwater development since the 1990 and the same is more pronounced in some of the districts like Bangalore, Belgaum, Chickballapura, Chitradurga, Kolar, Ramanagaram and Tumkur. Consequently, there was a steady decline in the water levels resulting in drying up of dug wells and shallow bore wells. The progressive farmers were trying their luck with deeper bore wells. Now it is common to have irrigation bore wells of the depth range of 250-300 m in parts of Kolar and Bangalore districts. The groundwater scarcity, decline in water levels, over- exploitation, deterioration in quality and related problems are encountered in the areas to the east of the Western Ghats covering the plateau region.

The ground water assessment was done on watershed wise. A total of 234 watersheds have been identified. These watersheds have been further classified as Command and Non Command as sub units. 86 of 234 watersheds are having command area and thus, a total of 310 assessment units are there. The Annual Replenishable Ground Water Resources has been assessed as 17.03 bcm and the Net Annual Ground Water Availability is 14.81 bcm. The Annual Ground Water Draft is 9.41 bcm and the Stage of Ground Water Development is 64%.

The resources so assessed have been apportioned on taluk wise for all the 176 taluks of the state. Out of 176 taluks, 95 taluks are having command area in parts. Thus, on a taluk wise basis there are 270 assessment units.

Out of 270 assessment units, 63 are categorized as Over-exploited, 21 as Critical, 34 as Semi-critical, and 152 are Safe. The over-exploited assessment units are concentrated in the eastern part of Belgaum and southern part of Bagalkote district, south eastern part of the State in the districts of Bangalore Rural, Bangalore Urban, Kolar, Chickballapur, Ramanagaram, Tumkur, Chitradurga and Davanagere. There has been marginal increase in the

annual replenishable resource in 2011 as compared to 2009 and there is a considerable decrease in draft. It is attributed to reduction in ground water abstraction structures mainly dug well which showed a decrease of 38%. However, numbers of borewells have been increased marginally about 2%. The reduction in number of dugwells is mainly attributed to drying of phreatic zone due to the over-exploitation in many of the districts, reduced draft in borewells due to mutual interference. This is cumulative effect of various contributory factors viz. reduction in yield due to mutual interference of bore wells in the OE area, reduction in draft due to the restriction in availability of electricity and farmers resorting advanced irrigation practices like micro irrigation as against the traditional flood irrigation especially in OE districts like Kolar.

KERALA

The state is underlain by diverse rock types of different geological ages from Pre-Cambrian to Recent. As much as 88% of the State is underlain by hard rocks and ground water is being tapped mostly through dug wells constructed in the weathered zone. Yields of open (dug) wells varies from 2 to 10 m³/day whereas that of bore wells ranges from less than 1 to 35 lps. About 12% of the State is underlain by Semi-consolidated and unconsolidated sedimentary formations where dug wells and filter points have yields of 1 to 35 m³/day, whereas deep tube wells have yields in the range of 1 to 57 lps. Laterites, which cover most of the geological formations in the major part of the state also forms an important aquifer in the state with dug wells having yields in the range of 0.5 to 6 m³/day.

The Ground water resources have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 6.69 bcm and Net Annual Ground Water Availability is 6.07 bcm. The Annual Ground Water Draft is 2.84 bcm and Stage of Ground Water Development is 47%. Out of 152 blocks, one block has been categorized as 'Over-exploited', two as 'Critical', 23 as 'Semi-Critical', 126 as 'Safe' and there are no saline block in the state. Comparatively high development of ground water has

been observed in Kasargod, Palakkad and Thiruvananthapuram districts. As compared to 2009 estimate, the Net Ground Water Availability and Gross Ground Water Draft as in 2011 have increased by about 0.73% and 0.95% respectively, resulting in the Stage of Ground Water Development remaining at about 47% in both the estimates. The changes in the recharge and draft parameters in respect of assessment units are due mainly to the reduction in the irrigation draft and modification of parameter values in the light of information gathered during subsequent field studies. Reorganization of blocks in 2010 has also resulted in change in category of ground water development in a few assessment units.

MADHYA PRADESH

The State of Madhya Pradesh has varied hydrogeological characteristics due to which ground water potential differs from place to place. The area is underlain by various geological formations ranging in age from the Archaean to the Recent. Hard rock areas cover more than 80% of total land area of the State. These hard-rock areas show wide variations and complexities in nature and composition of rocks, geological structures, geomorphological set up and hydro meteorological conditions. The crystalline rocks of Archaean age like granite, gneiss, granulites, schist, quartzite and granitoids occupy about 14.7% of geographical area of the State. The basaltic rocks of Deccan lava flows are the predominant formations and occupy nearly 44.5% of total geographical area. The consolidated sedimentary rocks of Vindhyan Super Group and Mahakoshal (Cuddapah) Super Group of Proterozoic age occupy about 19.1% of total geographical area and the semi consolidated (Gondwana Formation) occupies about 6.7%. Recent unconsolidated alluvial sediments occupy about 14.4% of total geographical area

Total recharge from rainfall in the State is of the order of 29,00,686 ham, (29.01 bcm) with Hoshangabad district having the highest recharge of 1.43 bcm ham and Alirajpur district has minimum recharge of the order of 0.19 bcm. Component of recharge from other sources is highest in Hoshangabad district

(0.71 bcm) followed by Dhar district (0.23 bcm) and Morena district (0.21 bcm) where maximum canal irrigation facility is available. Lowest value of recharge from other source is recorded in Annupur (355 ham) and Dindori (488 ham) being tribal districts where use of ground water as well surface water for irrigation purpose is very low. Thus, total recharge from rainfall is 29.01 bcm whereas from other sources is 6.03 bcm. The Annual Replenishable Ground Water Resources from all sources in the State is of the order of 35.04 bcm. The natural discharge during non-monsoon season in the State is 1.75 bcm. The net ground water availability in the state is 33.29 bcm.

The Annual Ground Water Draft of ground water for all uses in state is 18.83 bcm. Comparison of ground water draft for various uses reveals that draft for irrigation accounts for more than 93% of total ground water draft, where as draft for domestic & industrial supply accounts for meager 7% of the total ground water draft in the state. The stage of development of the State is 57%. Out of 313 blocks, 24 blocks has been categorized as 'Over Exploited', 4 as 'Critical', 67 as 'Semi-Critical' and 218 blocks as 'safe'.

The ground water resources of the individual block /assessment unit show wide variation in the resource available and stage of ground water development. In Madhya Pradesh, the ground water development concentrates mainly in the western (Malwa Region), Central (Narsinghpur districts), Budelkhand Region and part of Baghelkhand Region of the State. Almost all over-exploited blocks are falling in western part of Madhya Pradesh, which is known as "MALWA AREA" where ground water draft has increased many folds during past decades. The entire command areas of different blocks are falling under safe category. District wise analysis of data of ground water availability and annual ground water drafts indicate that two districts namely Indore and Ratlam are falling under "Over-Exploited" category where stage of ground water development is more than 100% as a whole.

The Annual Replenishable Ground Water Resources has increased in the 2011 estimate as compared to 2009. The reason being localized improvement in rainfall pattern, increased activity on rainwater harvesting and water conservation measures etc. In addition, management practices like efficient water use practices with community participation, increased awareness etc. also helped in improving water use efficiency of ground water resources in stressed areas. In majority of the cases, it is the combination of the above mentioned reasons which have brought in the changes in category.

MAHARASHTRA

The State is underlain by diverse rock types of different geological ages from Pre-Cambrian to Recent. Large part of the State is underlain by Basaltic hard rocks where dug wells are predominant. They mostly tap the weathered zone and fractures/joints. The yield of dug wells varies from 3 to 5 lps. However, in few district borewells are becoming more popular and they tap both shallow as well as deeper aquifers. A small part of the State is occupied by Semi-consolidated sedimentary rocks where tubewells have yield of 5 to 45 lps.

The Ground water resources have been assessed watershed wise and subsequently apportioned into block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 33.95 bcm and Net Annual Ground Water Availability is 32.15 bcm. The Annual Ground Water Draft is 17.18 bcm and Stage of Ground Water Development is 53%. Out of 353 blocks, 10 have been categorized as over-exploited, 2 critical, 16 semi-critical and remaining 325 as Safe. In Maharashtra, the ground water development is mainly concentrated in the central part of the state i.e. predominantly in the drought prone areas. The dependency on ground water is more due to non-availability of surface water. Hence, the over-exploited, critical and semi-critical blocks are mostly from the Central part of the State. Similarly, they have a direct correlation with the water guzzling crops.

As compared with 2008-09 estimates, there is decrease in Annual Replenishable Ground Water Resources in 2011 by about 5% which is mainly attributed to decrease in the recharge by water conservation structures. This is due to silting of the water conservation structures. The Ground Water Draft has increased by about 1.3% during the same period. This has resulted in increasing the number of over-exploited blocks from 9 to 10, critical blocks from 1 to 2 compared to 2008-09.

MANIPUR

The State of Manipur is occupied by mostly North South parallel hill ranges made up of consolidated and semi-consolidated rocks ranging in age from pre-Mesozoic to Miocene. The consolidated rocks confined to the eastern part of the state along the Myanmar border. The semi-consolidated formations, which cover almost the entire state, comprise shale, siltstone, sandstone and conglomerate. These formations belong to Disang, Barail, Surma and Tipam group of rocks. In the Western and central part of the state, unconsolidated alluvium of quaternary age occurs in the valleys and topographical lows. Groundwater is restricted to secondary porosity in joints, fissures, fractures and weathered residuum of consolidated and semi-consolidated rocks and inter-granular pore spaces of alluvial deposits. In the valley, ground water is utilized through tube wells, tapping granular zones of thickness ranging from 10 to 20 m, and the yield of the tube well varies from 10 to 30 m³/hr. The transmissivity and hydraulic conductivity of the aquifer formation range between 4.3 & 89 m²/day and from 0.67 & 16 m/day respectively.

Ground water resources have been assessed district-wise due to paucity of block wise data. The total area considered for the resources estimation is 2441.40 sq.km, which covers Imphal West, Imphal East, Thoubal, Bishnupur and parts of Churachandpur districts of Manipur valley. Rests of the four districts were excluded for the recharge re-assessment. The Annual Replenishable Ground Water Resource of the State has been estimated as 0.44 bcm and Net Annual Ground Water Availability is 0.40 bcm. The Annual

Ground Water Draft is 0.004 bcm and Stage of Ground Water Development is 1%. All the districts have been categorized as 'Safe' and there is no saline area in the state.

As compared to 2009 estimate, there is negligible change in annual ground water recharge (0.003 bcm) which may be due to low rainfall, especially in Manipur valley, during last three years.

MEGHALAYA

The Meghalaya State is essentially occupied by hard massive rocks belonging to the Archaean gneissic complex with acid and basic intrusives and Precambrian Shillong Group of parametamorphites. Groundwater occurs under unconfined condition in the weathered residuum and fractured rocks and restricted to about 150 m depth. The development of ground water is practiced by dug wells which is restricted to the weathered zone and through bore wells including hand pumps which mainly tap the semi-weathered and fractured zones in the hard rock. The south-western, southern and south-eastern parts of the state is covered by semi-consolidated formations comprising sandstones, shales, conglomerates, limestones etc. belonging to Cretaceous – Tertiary age. The aquifers are formed by rock strata that are granular/porous, fissured/fractured or cavernous. These aquifers are thick and discontinuous in nature. The unconsolidated sediments comprising sand, gravel, silt, clay, etc are found to occur as thin veneer along rivulets and as valley-fills. Significant thickness of this unconsolidated formation is found to occur only along extreme north-western fringe of the state in West Garo Hills district. Productive aquifer of thickness ranging of 10-15 m occurs in the area which is suitable for the development of ground water through shallow tube well.

Ground water resources have been assessed district-wise due to paucity of block wise/ watershed wise data. The Annual Replenishable Ground Water Resource of the State has been estimated as 1.78 bcm and Net Annual Ground Water Availability is 1.60 bcm. The Annual Ground Water Draft is 0.002

bcm and Stage of Ground Water Development is 0.08%. All the districts have been categorized as 'Safe'.

As compared to 2009 estimate, the Annual Replenishable Ground Water Resources as in 2011 has increased by 0.55 bcm. This change may be attributed to the fact that during 2011 estimation, assessment Area has been demarked based on digitized Geological and geomorphological maps and the area has increased for West Garo Hills District and East Garo Hills District. The changes in other figures are negligible.

MIZORAM

The state is occupied mainly by the rocks of the Tertiary formation ranging in age from Oligocene to Miocene to Recent. The Barails form the lower most rock units comprising siltstone and bands of soft and hard fine grained sandstone with strings of carbonaceous material and occur in the north eastern part of the state. The Surma is divided into two formations, Bhuban and Bokabil. The Bhuban is made up of grey sandstone and shale and occupies the major part of the state all along the length of the state. The Bokabil, predominantly argillaceous, mostly occurs along the western part of the state. The Tipam sandstone is of semi-consolidated in nature comprising medium to coarse grained sandstone with subordinate shale and occurs in limited extent in the north western part of the state. The alluvial deposits comprising silt, clay and sands occur in the valley fill area with very limited thickness. Ground water is confined only to valley filled areas and secondary porosities of semi-consolidated rocks. These aquifers are the main source for springs. Ground water stored in the hill slopes emanates in the form of springs, which are being used as a source for water supply. In the valley area, the yield potential of tube wells within the depth range of 200 m tapping Tertiary sandstone ranges from 120 to 330 liters per minute for drawdown of 13 to 20 m. The transmissivity, specific capacity and storativity are to the tune of 11 - 46 m²/day, 25.61 lpm/mdd and 4.28×10^{-4} respectively.

The major part of the State is covered with similar formations and hilly terrain with more than 20 % slope. However, ground water resources have been

assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 0.03 bcm and Net Annual Ground Water Availability is 0.027 bcm. The Annual Ground Water Draft is 0.001 bcm and Stage of Ground Water Development is 4%. All the blocks have been categorized as 'Safe' and there is no saline block in the state. As compared to 2009 estimate, there is no significant change in annual ground water recharge and ground water draft.

NAGALAND

The state is covered by rocks ranging in age from Pre-Cretaceous to Recent. The rock sequences comprise the geosynclinal facies, represented by Disang Group, Barail Group, Surma Group, Tipam Group, Namsang formation and Dihing Group. While the Disang and Surma Group of rocks are mainly argillaceous, the Barail and Tipam groups are arenaceous. The Girujan clay formation overlying the Tipam sandstones is characterized by typical blue, mottled clay and argillaceous sand stone beds. Older rocks occupy southern parts of the State whereas younger rocks are exposed in the northern parts. The unconsolidated alluvial plains, comprising clay, sand pebble, cobble and boulder assemblages, occupy the narrow, intermountain and open valleys in the northern part of the state bordering upper reaches of Brahmaputra flood plains of Assam. The consolidated formations are confined to the south eastern part of the State along the Burma (Myanmar) border.

Ground water development potentiality in valley fill and alluvial deposits are restricted to construction of open wells having depth of 15 to 20 metres and deep tube well down to 100 m depth which yield to the tune of 10 m³/day to 45 m³/day with more than 5 m drawdown. Water bearing formations pertaining to Tertiary deposits are found to have moderate potentials which can sustain deep tube wells having yield prospects varying from 10 to 20 m³/hr. The valleys underlain by Tipam sandstones form good aquifers with yield prospects varying from 30 to 80 m³/hr. In the consolidated formations, ground water abstraction structures can be constructed in structurally weak zones. Autoflow zones have also been identified in some parts of the state.

Ground water emerges as perennial springs which are the main source of water supply for domestic needs in the state.

The ground water resource estimation of the state is done on district-wise assessment unit due to paucity of block-wise data. The Annual Replenishable Ground Water Resource of the State has been estimated as 0.62 bcm and Net Annual Ground Water Availability is 0.55 bcm. The Annual Ground Water Draft is 0.03 bcm and Stage of Ground Water Development is 6.13%. All the districts have been categorized as 'Safe'. As compared to 2009 estimate there is an increase in annual ground water recharge by about 0.20 BCM in the 2011 estimate, which can be attributed to the non-monsoon recharge. The changes in other figures are negligible.

ODISHA

The state is underlain by diverse rock types, which ranges in age from Precambrian to Cenozoic era. The Precambrians occupy nearly 80% of the total geographical area of the State. The Tertiary and the Quaternary Alluvial formations are restricted mainly to the narrow coastal tracts. The Gondwana group of rocks belonging to Paleozoic and Mesozoic era occurs in isolated patches in different parts of the State. These formations occur in Talcher area of Angul district and Ib river valley area of Sambalpur and Sundargarh districts. Groundwater abstraction in the state is mostly done by dug wells constructed in the weathered zone in hard rock areas and in shallow phreatic aquifers in alluvial areas. The yield of open (dug) wells varies from 1 to 5 lps. However, at present, bore wells, shallow to medium deep tube wells, filter point tube wells are also in use for ground water abstraction both for domestic and irrigational purpose. The yield of bore wells varies from 2 to 5 lps in general depending on the occurrence of saturated fractures at depths. The yield from shallow and medium deep tube wells may vary from 6 to 10 lps in general depending on the aquifer disposition.

The Ground water resources in the state have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 17.78 bcm and Net Annual Ground Water Availability is

16.69 bcm. The Annual Ground Water Draft is 4.73 bcm and Stage of Ground Water Development is 28%. Among the total of 314 Blocks spanning across 30 districts of the state, 308 blocks have been assessed as safe, rest 6 blocks are completely saline/ poor ground water quality area. In Odisha, the ground water development is mostly concentrates in the eastern part of the state where agricultural usage is dominant.

As compared to the last assessment carried out during 2009, there is no significant change in the overall State level figures of dynamic ground water resources. However, significant changes have been observed in some of the blocks of the state. Based on extensive field studies, more reliable specific yield values were used. This has led to more realistic resource estimation in the current assessment than the Rainfall Infiltration Method which was adopted for majority of the assessment units in the previous estimation exercises.

PUNJAB

The Punjab State is a flat alluvial plain having a thin belt of mountains along north eastern border and stable sand dunes are seen dotting the landscape in the south western parts. The slope of the plain is 0.4 m/km towards south & southwest direction. The alluvial deposits in the State are comprised of sand, silt and clays often mixed with kankar. Sandy zones of varying grade constitute abundant ground water resources & act as a reservoir. The alluvial plain towards the hills is bordered by the piedmont deposits comprising Kandi and Sirowal. Immediately south-west of the hills, Kandi belt is 10 to 15 km wide followed by Sirowal which imperceptibly merges with the alluvial plain. Kandi deposit explored up to 450 m depth show gradation from boulders to clays and at places an admixture of various grades in different proportions. The Sirowal deposit is essentially composed of finer sediments but occasional gravel beds are also encountered in them.

Ground water levels in Punjab State vary from almost near surface to about 40 m bgl. The deep water levels are recorded in Kandi belt. Water logging conditions exist in some parts of south-western districts. In the remaining part of Punjab the water table varies from 3 to 34 m bgl. The

master ground water slope is towards southwest. Most of the shallow tubewells have ground water draft per year varying from 0.962 to 4.09 Ham (electric) and 0.712 to 2.11 Ham (diesel) and command area is 1 to 3 Hact. of land per tube-well in the State.

There is significant variation in quality of ground water with depth, especially in the south western parts of the State. As per recent estimates, the area underlain by ground water of unfit quality is around 3022 sq. km. (6 % of the total area of Punjab State).

The Dynamic Ground Water Resource of Punjab State has been assessed as per GEC-1997 Methodology by taking Block as a Unit of Assessment. At present, there are total 145 Blocks of Punjab State. The last estimation of Ground water Resources was carried out in 2009 for 138 Blocks. The Minor Irrigation Census for Tube wells for the year 2006-07 has been carried out by the Agriculture Department for 136 Blocks only. As such, 138 Blocks have been considered by incorporating the data of 2 Blocks on pro-rata basis, for making assessment of Dynamic Ground Water Resource.

The Ground water resources have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 22.53 bcm and Net Annual Ground Water Availability is 20.32 bcm. The Annual Ground Water Draft is 34.88 bcm and Stage of Ground Water Development is 172%. Out of 138 blocks, 110 block has been categorized as 'Over-exploited', 4 as 'Critical', 2 as 'Semi-Critical', 22 as 'Safe' and there are no saline block in the state. The block-wise stage of ground water development varies from 21% in DharKalan Block of Gurdaspur district to maximum of 416% in Ahmadgarh block of Sangrur district respectively. The district wise stage of ground water development varies from 69% in Muktsar district to 283 % in Sangrur district.

Shallow water level area having depth to water table less than 5 m bgl in the State is about 4584 km², which is lying mainly in the south-western districts of the Punjab State. Blockwise Annual Additional Potential Ground Water Recharge, has been worked out to be 1,204 Ham for water logged and

shallow water table areas of the State. The maximum potential has been observed in Muktsar District. In comparison with 2009, ground water resources assessment shows that one block has shown the change in the stage of development to higher category i.e. safe to critical.

As compared to 2009 estimate, the Net Ground Water Availability and Gross Ground Water Draft as in 2011 is almost same respectively, resulting in the Stage of Ground Water Development remaining at almost same in both the estimates. In general, the variation in stage of development has been observed which seems to be mainly due to increase in number of electric tubewells there unit draft and conversion of centrifugal pumps to submersible pumps.

RAJASTHAN

The State of Rajasthan has diversified geology, ranging from Archean metamorphics to recent alluvial sediments. Based upon geological diversities, geomorphological setup and ground water potentialities, the state of Rajasthan can be divided into three broad hydrogeological units.

(i) Unconsolidated formation (ii) Semi-consolidated formation (iii) Consolidated (Fissured formation). Large part of the State is underlain by Quaternary sediments (Thar Desert) consisting of clay, silt, sand and gravel of various grades. The fine sand and clay with or without kankar layers have formed multi layered aquifer system. The thickness may go up to 200-250 m below ground level, either resting directly over the basement rocks or over the Tertiary formation. Drilling data of CGWB indicates presence of aquifer of poor quality down to a depth of 200 mbgl. Exploratory drilling data reveals that the yield vary from meagre to 10 m³/day, transmissivity ranges between 80 and 300 m²/day and storage co-efficient vary from 1.1×10^{-5} to 3.9×10^{-6} in the state. In compact sedimentary rocks i.e. sandstone belonging to the Vindhyan formation is compact in nature and has low primary porosity. Ground Water occurs within the weathered residue and in the secondary porosity underneath. In general, the thickness varies from 5 to 10m. Yield potential is limited due to compact nature of the formation. The limestone is

also having low ground water potential. The yields of dug wells vary from 0.25 to 0.75 m³/day with water levels ranging from 10 to 120 meters. The yield of the wells drilled in Vindhayan formation has been observed to be 15m³/day, tapping fractures between 50-75 mbgl. In consolidated formation (Fissured) the thicknesses of the weathered zone vary from 5 to 50 m. Ground Water occurs under unconfined condition within the weathered zone. The results of the exploratory drilling carried out by CGWB in hard rock areas indicate presence of productive fractures down to a depth 40m and yield vary from 3 to 15 m³/day., whereas transmissivity vary from 3 to 30 m²/day.

The Ground water resources have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 11.94 BCM and Net Annual Ground Water Availability is 10.83 BCM. The total Ground Water Draft due to all sources is 14.84 bcm. Projected for 2025 is 1.89 bcm while the net groundwater availability for future irrigation development is only 0.91 bcm and the stage of ground water development in the state is 137 %. Out of 243 blocks, 172 blocks has been categorized as 'Over Exploited', 24 as 'Critical', 20 as 'Semi-Critical' and 25 blocks as 'safe'. 2 blocks has been categorized as 'Saline'.

SIKKIM

Sikkim is a small mountainous state characterized by rugged topography with series of ridges and valleys. The various rock types prevalent in the state are Pelitic, Carbonate rocks and Gondwanas over a gneissic basement. and occasional alluvial terrains along streams and river courses. Ground water occurs largely in disconnected localized pockets and in deeper fractures zones, springs are the main sources of water. The total discharge of the springs is about 0.044 bcm and its annual utilization for domestic purpose is about 0.0113 bcm. The stage of development is 26%. There is marginal decrease net annual ground water availability in the state in 2011 as compared to 2009.

TAMIL NADU

Tamil Nadu State is underlain by diverse hydrogeological formations, nearly 73% of the state is occupied by hard rocks, the semi-consolidated and consolidated formations are mainly confined in the eastern part including the coastal tract. In the hard rock area, groundwater is mainly developed through dug wells and dugcum bore wells tapping the weathered zone, the yield of open wells vary from 1 to 3 lps, whereas in dugwells tapping soft rocks including sedimentary formations, the yield is upto 10 lps.

The dynamic groundwater resources have been assessed firka wise. The Annual Replenishable Groundwater resource of the State has been estimated as 21.53 bcm and Net Annual Ground Water Availability is 19.38 bcm. The Annual Ground Water Draft is 14.93 bcm and Stage of Groundwater Development is 77 %. Out of 1129 firkas, 374 firkas has been categorised as 'Over Exploited', 48 as 'Critical', 235 as 'Semi-Critical' and 437 firkas as 'safe'. 35 firkas has been categorised as 'Saline'. Greater groundwater development is noticed in the central and western part of the state. The groundwater water draft is 14.93 bcm which is 10 % lesser than 2009 estimate. The overall stage of groundwater development of the state is 77 %. This is attributed to reduction in irrigation draft due to urbanisation in some regions of the state and marginal reduction in usage of dugwells for domestic use.

TRIPURA

The State is occupied by the rocks ranging in age from Upper Tertiary to Quaternary. Mobile trough geosynclinal deposition of Barail group followed by flysch type of Surma & Tipam sediments, overlain by Dupitila formation, is noticed in the state. Most of the longitudinal synclinal valleys of the state are the basins of deposition of recent formation. Recent alluvium occurs along the streams and the flood plains of major rivers.

Ground water occurs under unconfined condition in Dupitila formation, Recent formation & in Tipam formation. Besides it also occur under confined to semi-confined conditions in Tipam formation at considerable depth. Recharge areas for the deeper aquifer lies in the adjacent anticlinal hills.

Wherever a good thickness of impermeable clay beds underlie & overlie the saturated granular zones, autoflow artesian conditions have been found in the valleys, which are the discharge area. The artesian flowing conditions occur in patches both at shallow depth and at deeper depth. The auto discharge of the flowing wells in the State ranges from 100 to 6000 lph, the maximum auto discharge from deep tube well to the extent of 54000 lph has been found in Khowai valley near Khowai town, where the piezometric head rose up to 7m above ground level.

Ground water resources have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 2.59 bcm and Net Annual Ground Water Availability is 2.36 bcm. The Annual Ground Water Draft is 0.16 bcm and Stage of Ground Water Development is 7%. All the blocks have been categorized as 'Safe'. As compared to 2009 estimate, there is no significant change in annual replenishable ground water resources and annual ground water draft.

UTTAR PRADESH

The state of Uttar Pradesh is categorized with five distinct hydrogeological units – Bhabar, Terai, Central Ganga Plains, Marginal Alluvial Plain, Southern Peninsular area. Bhabar is mainly the recharge zone having deeper water levels. The ground water development in phreatic aquifer is through hand pumps, dug wells, dug cum bore wells and shallow tube wells. The yield from these wells has been generally found to be in the range of 40 to 60 lps. Terai zone lies between Bhabar in the North and Central Ganga Plain in the South. It is characterized by fine grained sediments with occasional pebbles and boulders. The average yield of tube wells constructed in this zone varies from 30 to 60 lps at moderate drawdowns. Central Ganga Plain constitutes the most promising ground water repository characterized by multi-layered aquifer systems. The yield of the open wells and hand pumps constructed in the phreatic aquifer vary from 5 to 10 lps. The tube wells in the phreatic aquifer yield between 20 and 28 lps at 6 to 8 m drawdowns. Marginal alluvial plain consists of kankar mixed clay-silt beds intercalated with sand and gravel

lenses. The aquifer in this area is capable of yielding 15 to 40 lps at moderate drawdowns. Southern Peninsular Region is characterized by sedimentary formations (sandstone, quartzite, limestone, shale). The wells tapping these formations generally recorded yield between 2 and 8 lps.

The assessment unit for ground water resources assessment in the state of Uttar Pradesh is block. The Annual Replenishable Ground Water Resource of the state has been estimated as 77.19 bcm and Net Annual Ground Water Availability is 71.66 bcm. The Annual Ground Water Draft is 52.78 bcm and Stage of Ground Water Development is 74%. Out of 820 assessment units (blocks), 111 have been categorized as Over-exploited, 68 as Critical, 82 as Semi-critical and 559 as Safe.

The estimates of Replenishable ground water resources have increased by about 2.58% as compared to the previous estimate of 2009. On the other hand, there has been a 6.66% increase in the ground water draft in the present estimate as compared to 2009.

UTTARAKHAND

Uttarakhand State has a distinct geological attributes with wide variety of rock units ranging in age from Archaean to Quaternary. About 85% of the geographical area of the state is mountainous and underlain by hard rocks. Ground water in the hard rock area is developed through the springs and hand pumps constructed in weathered zone. Discharge of springs in the Lesser Himalaya and Central Himalaya is variable and ranges from 60 to 600 LPM. About 15% of the geographical area is underlain by semi-consolidated and unconsolidated formations known as Tarai and Bhabhar. Ground water in this area is developed by open wells, shallow and deep tube wells. Unit draft of state owned deep tube wells varies from 15.08 to 25.80 ha m while that of shallow tube wells ranges from 1.91 to 2.20 ha m.

The ground water resources of Uttarakhand State have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 2.04 bcm and Net Annual Ground Water Availability is

2.00 bcm. The Annual Ground Water Draft is 1.13 bcm and Stage of Ground Water Development is 57%. Out of 18 blocks, no block has been categorized as 'Over-exploited'. Two blocks have been categorized as 'Critical', five blocks as 'Semi-Critical', eleven blocks as 'Safe' and there are no saline block in the State. Comparatively high development of ground water has been observed in Udham Singh Nagar and Haridwar districts. As compared to 2009 estimate, the Net Annual Ground Water Availability in 2011 has been decreased by 5.97 % while the Gross Ground Water Draft in 2011 has increased by about 8.09% respectively. This has resulted in increased Stage of Ground Water Development of 6% in 2011 compared to 2009 estimations. The changes in the recharge and draft parameters in respect of assessment units are due mainly to use of revised irrigation census, increased agricultural and industrial development in parts of the State. This has resulted in modification of parameter values in the light of information gathered during subsequent field studies.

WEST BENGAL

Nearly two third area of the state is occupied by unconsolidated sediments, the western part of the state is partly occupied by the hard rocks. The phreatic aquifer is generally developed through dug well, dug cum bore well and shallow tubewell. The yield of these wells varies from 1-5 lps.

Ground water resources have been assessed block-wise. The Annual Replenishable Ground Water Resource of the State has been estimated as 29.25 bcm and Net Annual Ground Water Availability is 26.58 bcm. The Annual Gross Ground Water Draft is 10.69 bcm and Stage of Ground Water Development is 40%. Out of 271 assessed units (blocks and Municipal areas), provisionally 53 have been categorized as Semi-critical and 01 as Critical and remaining 217 as Safe. There have been marginal changes in the estimation of ground water resources figures.

ANDAMAN AND NICOBAR ISLANDS

Nearly seventy five percent of the total geographical area of A&N islands is underlain by unproductive sedimentary group of rocks which is

developed only through dug wells having meager yield of 0.1-0.5 lps. 10% of the island area is occupied by igneous Ophiolite suite of rocks. Although restricted in occurrence, these formations are observed to yield moderate to high both in shallow and deeper locales and they are developed by dugwells and bore wells with yield ranging from 1-10 lps. The rest 15% of the island area is covered by Coralline Limestones associated Archipelago Group. This formation contains appreciable quantity of groundwater with yield varies from 5-25 lps. Ground water in the formations occurs under unconfined to confined condition.

Ground water resources have been assessed island-wise. The Annual Replenishable Ground Water Resources of the A&N Islands have been estimated as 0.308 bcm and Net Annual Ground Water Availability is 0.286 bcm. The Annual Gross Ground Water Draft is 0.013 bcm and Stage of Ground Water Development is 4.44%. Out of 36 assessed units (islands) all are categorized as Safe.

CHANDIGARH

Chandigarh is underlain by the alluvial deposits belong to Quaternary age and comprise layers of fine sand and clay. Coarser sediments occur along the Sukhna Choe and Patiali ki Rao whereas relatively finer sediments, thus restricting the aquifer disposition laterally, underlie the area between these two streams. Fair to good aquifer horizons occur in most part of Chandigarh comprising medium to coarse sand, to a depth of 180 mbgl below which they become finer. The yield of the deeper aquifers is also lesser as compared to the shallower ones. Ground water in the area occurs under water table, confined as well as semi-confined conditions. Ground water occurs under unconfined conditions down to about 80 m in Manimajra area. In other areas the semi-confined conditions prevail up to 20-30 m below land surface. The depth of the shallow aquifer system is less than 30 m below ground level whereas the depth of the deeper aquifer system ranges from 40 to 450 mbgl of explored depth. The transmissivity values for the deeper aquifer system ranged between 74 m²/day to 590 m²/day. The storativity

values ranged between 1.5×10^{-4} to 7.5×10^{-4} indicating confined nature of aquifer systems. The transmissivity values of shallow aquifers up to 100 m depth range obtained during these tests ranged between 70 and 466 m²/day. Ground water is found to be fresh and suitable for drinking as well as irrigation purposes. Normally, the ground water drawn from the deeper aquifers is less mineralized as compared to water drawn from shallow aquifers.

UT Chandigarh has very small area and whole UT has been taken as an assessment unit. Groundwater level data of the period 2006-10 has been used for estimation whereas water level trends of 2000-2010 have been used for long term trends for categorization. The Annual Replenishable Ground Water Resource of the UT has been estimated as 0.022 bcm and Net Annual Ground Water Availability is 0.019 bcm. The groundwater draft from unconfined aquifer is nil, since all the tube wells are deep and are drawing water from deeper aquifers. Since, there is no draft from shallow aquifers, the Chandigarh UT falls under SAFE Category. As compared to 2009 estimate, the net ground water availability as in 2011 is almost same.

DADRA & NAGAR HAVELI

Basaltic lava flows from the predominant aquifer system in the area. The ground water occurs under water table conditions in weathered basalts and vesicular basalts. The thickness of vesicular units ranges from 2 to 8 m. Ground water is developed by means of dug wells and dug cum bore wells. The sustainable yield of dug wells for 3 to 4 hours of pumping is 30 m³/day. The specific capacity of dug wells ranges from 3 to 300 lpm/m of drawdown in basalts. The transmissivity of shallow aquifer ranged between 5.5 to 305 m²/day. During 2010, the pre-monsoon depth to water levels in these wells ranged from 2.90 to 10.45 m.bgl, while the post-monsoon depth to water levels varied from 1.90 to 8.35 m.bgl. The depth to water levels during pre-monsoon interval during May 2011 varied from 2.35 to 12.70 m.bgl

In general the ground water resources assessment unit is Watershed, particularly in hard rock areas. In case of alluvial areas, administrative block

can also be the assessment unit. However, in this case considering the small area i.e., of 491 sq. km., entire UT of DNH is considered as a single unit for groundwater assessment purpose. The Annual Replenishable Ground Water Resource of the UT of DNH has been estimated as 0.062 bcm and Net Annual Ground Water Availability is 0.059 bcm. The Annual Ground Water Draft is 0.013 bcm and Stage of Ground Water Development is 22%. Since UT of DNH was considered as single unit (block/district), it has been categorized as 'Safe' and there is no saline block in the UT.

As compared to 2009 estimate, in 2011, the Net Ground Water Availability and Gross Ground Water Draft has increased by about 4.69% and 49.52% respectively, resulting in the increase in Stage of Ground Water Development from 15% in 2009 to 22% in 2011. The changes in the recharge and draft parameters is mainly due to increase in recharge from other sources and increase in irrigation and domestic drafts.

DAMAN & DIU

The Union Territory of Daman & Diu is underlain by alluvium and soft rock formation of Quaternary age and hard rock Basalt rock of Eocene – Late Cretaceous age. The entire island area of Diu is about 40 sq.km is underlain by alluvium & Milliolite soft rock formation. The Daman has about 72 sq km area out of which 30 % is covered by alluvium and the rest is underlain by Basalt rocks. In UT of Daman & Diu, dug well as well as dug cum bore wells are common for irrigation and domestic use. The yields of open (dug) wells varies from less than 1 to 5 m³/day whereas that of DCB wells ranges from less than 2 to 10 m³/day. The deeper aquifer zones have brackish to saline groundwater causing groundwater quality constraint for the development in UT of Daman & Diu. Due to limited thickness of good quality aquifers and relatively high development, the Diu district (Island) is assessed Over Exploited. Similarly, the Daman district also has high development but its water resources are augmented by Surface water (canal water) for irrigation and also for domestic usage.

The Ground water resources have been assessed on district wise basis. The Annual Replenishable Ground Water Resource of the UT of Daman & Diu has been estimated as 0.018 bcm and Net Annual Ground Water Availability is 0.0169 bcm. The Annual Ground Water Draft is 0.016 bcm and Stage of Ground Water Development is 97 %. Out of 2 assessment units (District area), the Diu is categorized as 'Over-exploited' and the Daman as 'Semi-Critical'. As compared to 2009 estimate, the Net Ground Water Availability have increased by about 43% whereas Gross Ground Water Draft have also increased by 49% as on 2011 and therefore, resulting in the decrease in Stage of Ground Water Development from 99% in 2009 to 97% in 2011. The changes in the recharge and draft parameters in respect of assessment units are due to mainly increase in recharge activities and increase in rainfall during the last five years leading to modification of parameter values.

LAKSHADWEEP ISLANDS

Lakshadweep islands are composed of calcareous sand and materials derived from coral atolls. Alternate layers of loose sand, moderately cemented calc-arenites and well cemented, hard and compact limestone underlie the islands. In these islands fresh groundwater occurs under phreatic conditions as a lens floating over the saline water and is in hydraulic continuity with sea water. Water levels in wells are strongly influenced by tides. Dug wells are the common ground water abstraction structures in the islands.

Dynamic ground water resources have been assessed for individual islands by computing various components of recharge and draft. Rainfall is the only source of recharge in the islands, whereas domestic draft, evapotranspiration losses and water loss due to outflow into the sea are the major components of draft. A part (20%) of the annual water surplus is kept as reserve to accommodate delayed or deficit monsoon years.

As per the assessment, the total annual ground water surplus in the islands amounts to 1054 Ha.m (0.011bcm). The annual ground water draft, for all the islands grouped together, amounts to 234 ha.m (0.0025 bcm). The stage of

ground water development for the group of islands is of the order of 67%. Based on the stage of ground water development, Agatti, Amini and Kavaratti islands have been categorized as 'Semi-critical' and the remaining islands as 'Safe'. A comparison with the earlier estimate of 2009 indicates a reduction in the draft component in all the islands as the actual population during 2011 was found to be less than the projected population considered for the 2009 estimation based on the decennial growth rate (1991-2001). As a consequence, the domestic draft has decreased by about 10% for the group of islands as a whole. This has resulted in improved ground water balance and consequently, lower stage of ground water development in all the islands.

PUDUCHERRY

The UT of Puducherry is underlain by the semi-consolidated and unconsolidated sedimentary formations which mainly sustains dug wells and shallow and deep tube wells. The yield of the wells generally varies between 3 to 15 lps. High yielding wells in the range of 10 to 40 lps exists in the Tertiary sandstones. The Dynamic groundwater resources for UT of Puducherry have been assessed Region wise.

The Annual Replenishable Groundwater resource of the UT of Puducherry has been estimated as 0.189 bcm and Net Annual Ground Water Availability is 0.170 bcm. Out of 4 regions, 1 region has been categorised as 'Over Exploited' and 2 regions as 'safe'. 1 region has been categorised as 'Saline'. The groundwater draft is 0.15 bcm and the overall stage of groundwater development of UT of Puducherry is 90 %. The re-assessment of Dynamic Groundwater resources of UT of Puducherry in 2011 has shown marginal increase in the annual replenishable ground water resource as compared to 2009 estimate. Puducherry region has the highest stage of development (139%).

5.6 COMPARISON BETWEEN 2009 ASSESSMENT AND 2011 ASSESSMENT

The overall country figures of ground water resources assessment carried out in 2011 do not show any significant change as compared to 2009 estimate

except for the fact that the number of assessment units have increased by 765 no. (about 13%) from the previous assessment. The main reason for changes in the number of assessment unit is that previously (till 2009), assessment in Tamil Nadu was carried out block-wise basis. In 2011, the assessment unit is Firka which is a smaller administrative unit than block. Hence number of assessment units in Tamil Nadu has increased considerably. At the state level, however, noticeable changes in the assessment figures have been observed due to various reasons.

Rationalization of ground water recharge and draft norms and strengthening of database is the reason behind changes in resources estimation in case of most of the states like Assam, Goa, Himachal Pradesh, Karnataka, Maharashtra, Uttarakhand, West Bengal, Daman & Diu etc. States like Andhra Pradesh, Gujarat show increase in recharge and decrease in draft in some parts due to above average rainfall in recent years. Increase in recharge estimates in some of the assessment units in the states of Gujarat and Karnataka are also attributed to effective implementation of water conservation measures. The increase in ground water draft and consequent rise in Stage of Ground Water Development and Category is attributed to the growth of ground water structures to meet food security and domestic water needs of the rising population. The detailed reasons for significant changes in resources assessment figures between 2009 and 2011 estimates are given in **Appendix G.**

CHAPTER 6

CONCLUSION AND RECOMMENDATION

The annual replenishable ground water resources of the country have been reassessed as 433 bcm as on March, 2011. The ground water resources get replenished through rainfall and other sources like return flow from irrigation, canal seepage, recharge from water bodies, water conservation structures etc. The main source of replenishable ground water resources is recharge through rainfall which contributes to nearly 68% of the total annual replenishable resources. The net annual ground water availability of the country has been assessed as 398 bcm after keeping a provision for natural discharge. The annual ground water draft of the country as on March, 2011 is 245 bcm, the largest user being irrigation sector (about 91%). The stage of ground water development for the entire country, which is the percentage of ground water draft with respect to net annual ground water availability, has been computed as 62%. The utilization pattern of ground water is however uneven across the country resulting in ground water stressed conditions in some parts of the country while in other areas, ground water utilization have been sub-optimal. In 2011, assessments were carried out in 6607 assessment units (blocks/mandals/taluks) which is a significant increase from the number of units assessed in 2009 (5842 units). This shows a gradual refinement in the dynamic ground water resource assessment (as on March, 2011) as compared to previous assessments. Out of the total assessment units, around 16% are categorized as Over-exploited. The critical units are 3% of the total assessment units, semi-critical around 11% and safe are 68%. The remaining units (around 1%) are having saline ground water. The over-exploitation of ground water resources are caused by various region-specific reasons which are broadly spread over three different parts of the country. The assessment units located in the north-western part of the country in the states of Punjab, Haryana, Delhi and western Uttar Pradesh have plenty of replenishable ground water resources but because of unregulated extraction beyond the ground water recharge limits, most of these units are Over-exploited. Over-

exploited units are also common in western part of the country particularly in Rajasthan and Northern Gujarat adjacent to Rajasthan because of arid climate resulting in less recharge of ground water and hence stress on the resource. In peninsular India, over-exploited units are widespread in the states of Karnataka, Tamil Nadu and parts of Andhra Pradesh which is mainly because of lesser availability due to the poor aquifer properties of the hard rock terrains prevalent in the region.

The annual replenishable ground water resources of the country as a whole do not indicate any appreciable change from previous assessment as on March, 2009. However there have been significant variations at the scale of assessment units (blocks / watershed) at several places mainly because of – localized changes in rainfall pattern during last few years, ground water management interventions and improved database resulting in refinements in assessments.

Recommendation:

The Ground Water Resources Assessment of the country is being carried out at regular interval with the objective to identify and prioritize the areas for ground water management interventions. Considering the significance of this exercise in identification of areas for ground water management, continuous endeavor is required to bring in refinements in assessment. Some important issues which require immediate attention and deliberation are enumerated below.

a. *Aquifer Characterization and parameter estimation:* One of the key elements in ground water resources assessment is the estimation of the recharge and discharge parameters. It is recommended that experimental studies to be taken up in National Project on Aquifer Management (NAQUIM) on Rainfall Infiltration Test (for refining the norms of RIF and return flow from irrigation based on soil types and agro-climatic zones), Aquifer Performance Test (for determination of specific yield), real time monitoring of water level along rivers and canals (for studying the inflow-outflow pattern and consequent recharge-discharge rate from streams/canals), unit draft study

and random sampling of wells (for refining the norms and well census figures related to ground water draft assessment).

b. *Scale of assessment:* In the recent assessments, States like Andhra Pradesh and Tamil Nadu have adopted smaller administrative units viz. Mandal (Andhra Pradesh) and Firka (Tamil Nadu) as ground water assessment units. Even Karnataka is also carrying out ground water assessment on a larger scale. The reason for assessment at a larger scale by these States is mainly to introduce more pragmatic ground water management interventions. It is recommended that other states should also gradually attempt for micro-level assessment.

c. *Water Balance Studies:* Ground water is one of the several components of the Hydrologic Cycle, other important components being rainfall, surface water, soil moisture and evapotranspiration. Holistic water resources management interventions require proper understanding of the interaction between the different components of the hydrosphere. Therefore purpose driven studies on Water Balance of various watersheds need to be taken up –
a. For judicious management of the water resources, b. To study the climate change effects on the Water Resources.

d. *Case studies linking assessment with management:* As mentioned earlier in the chapter, the ground water assessment is being carried out to facilitate ground water management programmes. It is recommended that few case studies in various assessment units may be undertaken wherein quantitative evaluation of the ground water management interventions and consequent changes in the assessment results would be analyzed. Such studies would help in bringing out the efficacy of various management programmes and also the confidence level of the estimation.

e. The frequency of assessment at present is biannual. However it is felt that the frequency should be reduced to once in 5 years since there are not much significant short term changes also this is a very large scale exercised. The bench marking committee of international experts set up for technical upgradation of CGWB has also recommended for the assessment to be done once in 5 years.